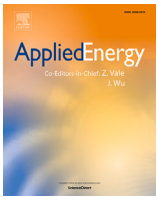
Applied Energy 393 (2025) 126095



Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apen





Data center load modeling through optimal energy consumption characteristics: A path to simultaneously enhance energy efficiency and demand response quality

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HIGHLIGHTS

- A hierarchical load modeling framework for data center demand response is proposed.
- Both real-time interactive workloads and latency-tolerant batch jobs are considered.
- A weakly nonlinear load model for demand response through energy optimization is proposed.
- Four optimal energy consumption patterns of data centers have been characterized.
- Both the energy efficiency and demand response quality of data centers have improved.

ARTICLE INFO

Keywords:
Data center
Demand response
Demand response quality
Energy efficiency
Load modeling
Optimal energy consumption characteristics

ABSTRACT

In an era defined by the rapid advancement of artificial intelligence and the global pursuit of "carbon neutrality," data centers face the dual challenge of enhancing energy efficiency while ensuring high-quality participation in power system demand response. However, conventional linear load models used in demand response programming often force data centers into a trade-off: sacrificing energy efficiency to ensure response quality, or vice versa. This paper presents a hierarchical load modeling framework that captures the optimal energy consumption characteristics of data centers to mitigate this conflict. At the foundational layer, a fine-grained, cross-system energy consumption model is developed to capture the intricate electrical-thermal-performance interactions among the computing, cooling, and power conditioning systems within the data center. Solving the energy optimization problem at this layer yields the optimal energy consumption characteristics of the data center. At the upper layer, these characteristics are analyzed and abstracted into a weakly nonlinear demand response-oriented load model, composed of four patterns that together form a piecewise function—two linear and two nonlinear regions—each corresponding to distinct workload conditions. The nonlinear relations are simplified from cubic to quadratic forms without significant loss of accuracy. Experimental results show that the linear regions achieve $R^2 \geq 0.9999$ with mean relative errors below 0.1404 %, while the quadratic regions reach $R^2 \geq 0.9982$ with mean relative errors under 0.6259 %. Applied to a typical demand response program, the proposed model reduces electricity costs by 13.40 % to 30.21 %, energy consumption by 24.19 % to 38.31 %, and cumulative curtailment deficit by 98.09 %, compared to conventional linear models.

1. Introduction

As the cornerstone supporting the development of digital and intelligent technologies, data centers are experiencing continuous growth in both number and individual size, consuming a significant amount of electricity. It is estimated that their operational energy consumption accounts for about 1 %–2 % of the current global total

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https://doi.org/10.1016/j.apenergy.2025.126095
Received 26 January 2025; Received in revised form 18 April 2025; Accepted 8 May 2025
Available online 19 May 2025
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electricity consumption [1–3], and this figure is expected to reach 3 %–13 % by 2030 [4]. Meanwhile, with the advancement of the "carbon neutrality" goals, there has been rapid and sustained growth in the demand for low-cost demand response resources in the power system [5]. Considering the potential for load scheduling in both space and time, data centers are seen as a kind of excellent demand response resource that can play an important role in addressing future challenges in power systems [6]. Against the dual background mentioned above, how to enhance the energy efficiency of data centers while simultaneously providing high quality flexibility resources for the power system has become a widely concerned issue among both the academic and industrial communities.

To enhance energy efficiency, data centers can implement measures such as replacing equipment with high-efficiency models, retrofitting building envelopes, and improving operational efficiency. Energy management optimization techniques, exemplified by dynamic voltage and frequency scaling (DVFS) [7] and dynamic thermal management (DTM) [8], serve as the primary methods for enhancing energy efficiency in data centers at the operational level [9]. With the advancement of research, these approaches are becoming more refined and systematic. For example, Wan et al. [10] proposed a cross-layer optimization framework for data centers that tracks workload arrival rate and collaboratively optimizes system parameters in chip, server, and computer room layers (including cooling system) to reduce data center energy consumption. Zhou et al. [11] proposed a cross-system modeling and optimization framework for direct air-side free cooling data centers. This framework considers the nonlinear characteristics, including those related to computational performance, energy consumption, and heat transfer, to optimize the holistic energy consumption of computing, power supply and distribution, and cooling subsystems within the data center. Specifically, it collaboratively optimizes the number of active servers, the status of the cooling system supply air (including fresh air ratio, temperature, humidity, and flow rate), the flow rate of chilled water, and the cooling water supply and return temperatures to adapt to changes in the arrival rate of computational workload and outdoor environmental temperature and humidity conditions. The range of objects and system levels considered in data center energy management is expanding, while the nonlinear characteristics of components and systems are increasingly factored into optimization problems.

In demand response, data centers regulate terminal power consumption patterns through various methods, including but not limited to [12,13]: (a) migrating and shifting computational workload spatially (across multiple geographically distributed data centers) and temporally (within the data center); (b) adjusting the operational parameters of systems and devices within the data center; and (c) controlling the charging and discharging of the battery in uninterruptible power supply (UPS) units. These adjustments involve not only temporarily reducing electricity consumption during peak load periods in the power system but also temporarily increasing consumption during times of excess renewable energy generation. Besides, the demand response should be based on flexibility rather than energy waste. Regarding participation mechanisms, data centers can engage in demand response through either price-based or incentive-based approaches. Specifically, the former includes time-of-use (TOU), critical peak pricing, and real-time pricing, while the latter encompasses direct load control, interruptible/curtailable service, and demand bidding/buy back, etc. [14]. Regardless of the method employed, an accurate data center load model is needed to address questions about the power load intensity that devices and systems can respond to, the duration of sustained responses, and the implementation of these responses. This is vital for data centers to participate effectively in demand response.

The computational complexity of large-scale optimization problems in demand response programs is significant. Therefore, the sophisticated nonlinear models commonly used for data center energy optimization are computationally intractable in demand response scenarios. Current research often simplifies data center load models into linear representations, formulating the associated optimization problems as Mixed-Integer Linear Programming (MILP) problems [15]. Table 1 provides an overview of the load models employed in studies of data centers involved in demand response over the past decade (since 2015) [16–32]. Specifically, the computing power within data centers is typically simplified as a linear model based on server utilization [16–32]. The cooling power is calculated by dividing the power of the information technology (IT) system by a fixed coefficient of performance (CoP) value

Table 1
Load models used in research on data center demand response over the past decade (since 2015).

Ref.	Workload type		Server/IT/Computing system					Cooling system	PCU
	Interactive workload	Batch jobs	Computing power	Leakage power	Server fan power	Chip temperature threshold	QoS	Cooling power	Power loss
[16]	✗	✓	Linear with utilization	✗	✗	✗	✗	Considered in a fixed PUE	
[17–19]	✗	✓	Linear with utilization	✗	✗	✗	✓	Constant CoP	✗
[20]	✓	✗	Linear with utilization	✗	✗	✗	✗	Considered in a fixed PUE	
[21–24]	✓	✗	Linear with utilization	✗	✗	✗	✓	Considered in a fixed PUE	
[25,26]	✓	✓	Linear with utilization	✗	✗	✗	✓	✗	✗
[27]	✓	✓	Linear with utilization	✗	✗	✗	✓	Considered in a fixed PUE	
[28,29]	✓	✓	Linear with utilization	✗	✗	✗	✓	Constant CoP	✗
[30]	✓	✓	Linear with utilization	✗	✗	✗	✓	Linear with IT power	✗
[31]	✓	✓	Linear with utilization	Linear with server inlet temperature	✗	✗	✓	Constant CoP	✗
[32]	✓	✓	Linear with utilization	✗	✗	✗	✓	Considered in the variable PUE	
This work	✓ (Throughout the model)	✓	Nonlinear (DVFS) (Eq. 6)	Nonlinear (chip temperature) (Eqs. 7–8)	✓ (Eq. 9)	✓ (Eqs. 16–19)	✓ (Eqs. 1–5)	Nonlinear (Eqs. 14–15)	Nonlinear (Eqs. 11–12)

for the cooling system [17–19,28,29,31]. There are also some studies [16,20–24,27] that use a method based on the power of the IT system and a fixed or variable power usage effectiveness (PUE) value to calculate the power of other systems besides the computing system. Compared to the continuous refinement and systematization of models used to characterize energy consumption patterns in data center optimization research, load models applied in demand response are often overly simplified. For example, they neglect DVFS, a chip-level technique for optimizing energy efficiency and computational performance. Additionally, critical aspects such as chip temperature thresholds and their influence on leakage power are typically omitted. Moreover, the energy consumption and nonlinear behaviors of cooling systems and power conditioning units (PCUs), which constitute the two other primary energy consumers in data centers apart from computing systems, remain insufficiently accounted for.

While these simplifications improve the mathematical tractability of large-scale optimization problems in demand response programs, excessive simplification may result in models that fail to accurately capture the physical realities of systems and devices within the data center. Such limitations may lead to unrealistic results and misleading conclusions in demand response programs, causing data centers to miss significant opportunities for enhancing energy efficiency or diminishing their performance during participation in demand response. Specifically, when a data center receives a computational workload allocated based on demand response programming, it encounters a dilemma. If one wishes to ensure the quality of data center participation in power system demand response, operations must conform to the (linear) patterns delineated by the load models. This often results in unnecessary energy consumption, which may even prove unfeasible. If one aims to sustain high energy efficiency in data centers, their energy consumption characteristics will differ from those of the currently oversimplified load models, potentially degrading the quality of demand response. As a result, data centers face challenges in enhancing energy efficiency while providing high-quality demand response to the power system when handling computational workloads allocated through demand response programming.

In summary, at the modeling level, a conflict exists between improving data center energy efficiency and enhancing their engagement in grid demand response. For one thing, to improve data center energy efficiency, intricate data center load models incorporating many nonlinear factors are developed, which make their computation intractable in demand response optimization. For another, to enhance their engagement in demand response, data centers often rely on oversimplified linear load models for computational tractability, resulting in deviations from optimum and reduced energy efficiency. To overcome this limitation, this study proposes and validates a hierarchical data center load modeling framework that effectively reconciles the need for model refinement to enhance energy efficiency with the requirement for model simplification to ensure computational tractability in demand response applications. This is achieved in two steps: first, energy consumption is optimized through fine-grained, cross-system models that exhibit strong nonlinear characteristics; second, weakly nonlinear load models are developed based on the external load characteristics revealed by the optimization solutions. The main contributions of this work include the following:

(1) A data center load modeling framework based on optimal energy consumption characteristics is proposed. This framework employs a hierarchical structure that balances the need for model refinement to optimize energy efficiency with the demand response requirement for computational complexity. (A brief comparison with existing studies can be found in Table 1.)
(2) Optimal energy consumption characteristics at both the module- and data center-level are analyzed and extracted. Based on these characteristics, a simplified load model with weak nonlinearity for data center demand response is developed, and the resulting losses due to simplification are evaluated.
(3) The performance of data centers using conventional linear load models and simplified nonlinear load models in demand response programming is evaluated and compared from the perspectives of electricity costs, energy consumption, and demand response quality, thereby validating the efficacy of the proposed load modeling framework.

The structure of the remainder of this work is as follows. Section 2 introduces the methodology of this study. In Section 3, the system configuration of the hypothetical data center addressed in this study is described. Additionally, the cross-system model and optimization problem at the foundational layer for enhancing energy efficiency are introduced and formulated, respectively. Following the solutions of the energy optimization problem, Section 4 presents the optimal energy consumption characteristics of the data center from both the module- and data center-levels, and proposes a simplified data center-level load model for demand response, characterized by weaker nonlinearity. This section also introduces a representative procedure through which a group of data centers participates in power system demand response. In Section 5, the effectiveness of the proposed nonlinear load model is validated through comparative experiments under the typical demand response scenario, demonstrating its superiority over conventional linear load models. Section 6 outlines the conclusions drawn from this study.

2. Methodology

Fig. 1 illustrates the methodological framework of this study, which encompasses foundational and upper layer modeling, along with experimental validation and analysis. In the foundational layer of the framework, a fine-grained, cross-system energy consumption model is developed to capture the intricate electrical-thermal-performance interactions among the computing, cooling, and power conditioning systems within the data center. Then, considering constraints such as QoS, chip temperature limits, device capacity, and thermal environment requirements, a data center optimization problem that optimizes several operational parameters—including computational workload allocation, chip frequency, server fan speed, and the temperature of the cooling air supplied by the computer room air conditioning (CRAC) unit—are proposed, which can be solved by heuristic algorithms (such as genetic algorithms). Subsequently, energy optimization is conducted under various computational workload conditions to extract the optimal energy consumption characteristics of the data center. It is worth noting that although this process involves substantial computation—mainly due to the diversity of workload conditions—it is specifically designed to establish the optimal energy consumption characteristics for the data center. Therefore, the time required to solve these problems does not impose any computational burden on the solution process of the demand response scheduling problem. Theoretically, this computation is a one-time effort and does not need to be repeated unless major changes occur, such as infrastructure reconfiguration or updates to computing performance constraints. Once the optimal characteristics are obtained, no further computation is necessary.

In the upper layer of the modeling framework, the solutions to the energy optimization problem at the foundational layer under various computational workload conditions are analyzed to extract the optimal energy consumption characteristics of the data center. Specifically, these characteristics illustrate the relationship between computational workload conditions and optimal (most energy-efficient) operating parameters of the system, as well as the relationship between computational workload conditions and optimal energy consumption. Given the modular trend in data center development, the analysis and extraction of optimal energy consumption characteristics encompass both the module and data center levels. Based on these characteristics, a simplified load model that meets the computational complexity requirements for data center demand response is established. Distinct from the conventional

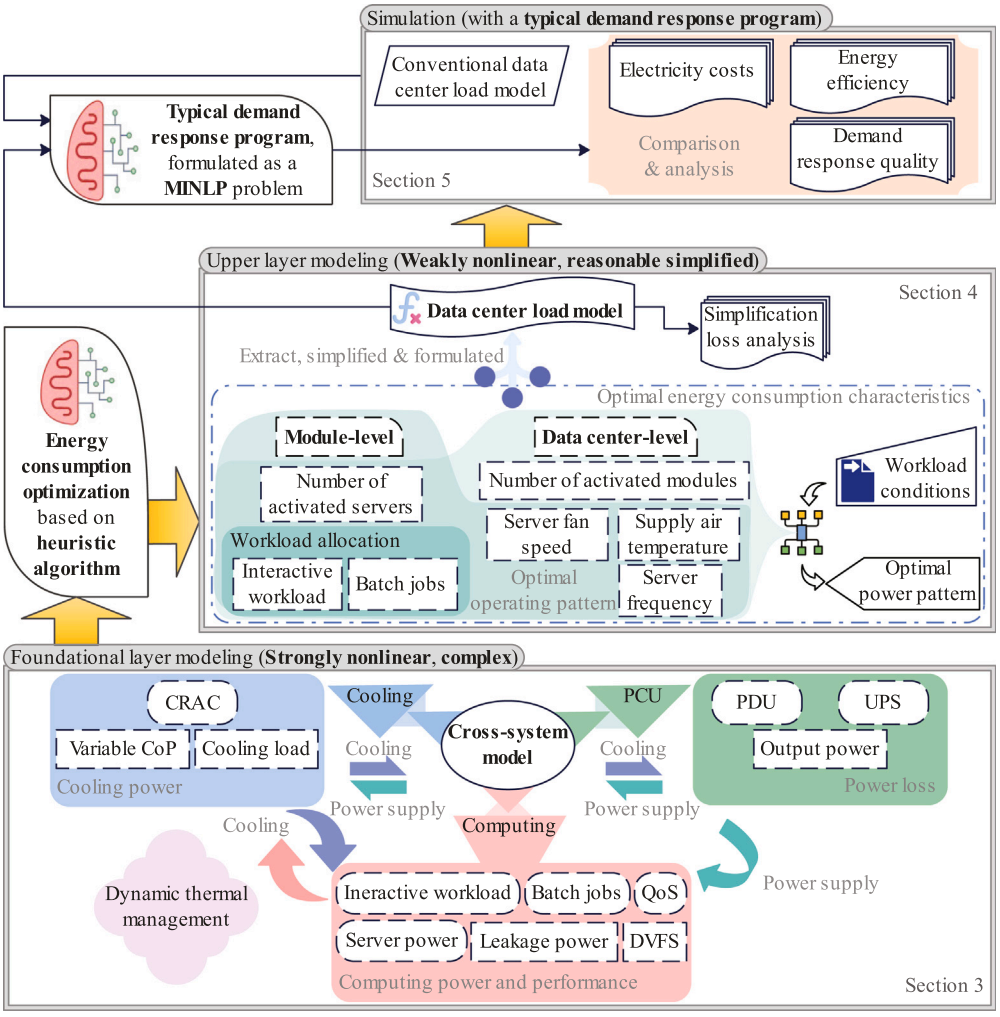


Fig. 1. Methodological framework.

load model, this model is expected to be a weakly nonlinear, compatible with energy efficiency improvement, and reasonably simplified model. An evaluation of the losses resulting from the model simplifications is also conducted.

Finally, in the simulation experiment, the data center load model obtained from the upper layer modeling and the conventional data center load model are embedded into a typical demand response program, formulated as a mixed-integer nonlinear programming (MINLP) problem, to compare and validate the effectiveness of the modeling approach proposed by this study. Specifically, these comparisons are conducted from three aspects: electricity costs, energy consumption, and demand response quality.

3. Foundational layer modeling for energy efficiency improvement

In this section, the system configuration of a hypothetical data center addressed by this study is introduced first. This section also presents the cross-system model of the data center and provides a mathematical formulation of the energy consumption optimization problem in the foundational layer modeling.

3.1. System configuration

Given the intricate network of diverse systems and equipment within data centers, where the array of different types of equipment complicates accurate systemic representation, modeling and analysis of individual components are insufficient to capture the reality of the entire system. Consequently, this study constructs a Tier IV hypothetical data center composed of an extensive array of equipment to further the research.

The setup encompasses 2000 servers distributed among 20 modules, with each module housing 20 racks. In compliance with the Tier IV data center requirements for power distribution and cooling equipment redundancy [34], which stipulate a 2(N + 1) configuration, each module is outfitted with two UPS units, two power distribution units (PDUs), and two computer room air conditioners (CRACs), one for use and one for backup. Furthermore, the data center includes additional sets of two UPSs, two PDUs, and two CRACs to meet redundancy requirements. Table 2 details the specific configuration of the system.

3.2. Cross-system modeling of data center module

In modular data centers, each module typically encompasses the essential components—computing (servers), cooling, and power conditioning facilities—forming the smallest complete unit of the data center. Consequently, this section describes the development of the cross-system model for data centers through the modeling of these individual modules. The focus is on three key modeling aspects: computational performance, power consumption (including computing power, cooling power, and power loss by PDUs), and thermal performance.

Table 2
System configuration of the hypothetical data center.

Facility	Specifications and Refs.	Configurations
Chip	Intel Xeon Platinum 8380 processor, with a thermal design power of 270 W, and a maximum allowable temperature of 83 °C [35,36]	2 chips/server
Chips heatsink	High performance heatsink [36,37]	1 heatsink/chip
Server fan	According to Ref. [38], with a maximum speed of 9000 rpm, and a rated power of 5.64 W	6 units/server
Server	HPE ProLiant DL380 Gen10 server, with full load of 619 W [39]	5 servers/rack
Rack	Equipped with 5 servers	20 racks/module
PDU	According to Ref. [40]	2 units/module, one for use and one for backup, and another 2 units for backup at the data center level
UPS	According to Ref. [40]	
CRAC	According to Ref. [41]	
Module	Equipped with 20 racks, 2 PDU, 2 UPS, and 2 CRAC unit	20 modules/data center

3.2.1. Computational performance

Generally, the active servers in a data center module can be categorized into three types: Type A servers that exclusively handle interactive workloads, Type H servers that handle hybrid tasks involving both interactive workloads and batch jobs, and Type B servers that exclusively handle batch jobs. Enabled by virtualization [42,43], it is assumed that different types of workloads are evenly distributed on the same type of active servers. Then, the workload allocated to different types of active servers within a data center module can be calculated as follows:

$$\lambda^{\mathrm{a}} = \frac{a^{\mathrm{a}}}{N^{\mathrm{a}}}, \tag{1}$$

$$\lambda^{\mathrm{h}} = \frac{a^{\mathrm{h}} + b^{\mathrm{h}}}{N^{\mathrm{h}}} = \frac{a - a^{\mathrm{a}} + b - b^{\mathrm{b}}}{N^{\mathrm{h}}}, \tag{2}$$

$$\lambda^{\mathrm{b}} = \frac{b^{\mathrm{b}}}{N^{\mathrm{b}}}, \tag{3}$$

where, a and b respectively represent the interactive workload and batch jobs allocated to a data center module, in unit/s; a^{a} and a^{h} represent the interactive workload allocated to Type A and Type H servers within a data center module, in unit/s, respectively; b^{h} and b^{b} are the batch jobs allocated to Type H and Type B servers within the module, in unit/s, respectively; λ^{a}, λ^{h}, and λ^{b} represent the arrival rates of workload allocated to a single active Type A, Type H, and Type B server within a data center module, in unit/s, respectively; N^{a}, N^{h}, and N^{b} respectively indicate the number of active Type A, Type H, and Type B servers in a data center module, in unit.

Assuming that the arrival of real-time interactive workload follows a Poisson process (an assumption widely used in other studies [10,44,45]), and that real-time interactive workloads are given priority execution in Type H servers. The expected response time for processing real-time interactive workload by Type A and Type H servers within a data center module (denoted as $t_{\mathrm{D}}^{\mathrm{a}}$ and $t_{\mathrm{D}}^{\mathrm{h}}$, in s, respectively) can be calculated using the M/M/1 queueing theory model as follows:

$$t_{\mathrm{D}}^{\mathrm{a}} = \frac{1}{f^{\mathrm{a}}\mu - \lambda^{\mathrm{a}}}, \tag{4}$$

$$t_{\mathrm{D}}^{\mathrm{h}} = \frac{1}{f^{\mathrm{a}}\mu - \frac{a - a^{\mathrm{a}}}{N^{\mathrm{h}}}}, \tag{5}$$

where, $\mu = 100$ unit/s represents the service rate of the server at its rated operating frequency [10]; $f^{\mathrm{a}} \in [0,1]$ and $f^{\mathrm{h}} \in [0,1]$ represent the relative value of the operating frequency of Type A and Type H servers, in dimensionless, respectively.

3.2.2. Computing power

The power related to computing in servers can be divided into three parts [33,46,47]: idle power, dynamic computing power, and leakage power. As shown in Eq. (6), this study uses a model [10] based on operating frequency and utilization of the chip to calculate the sum of idle and dynamic computing power consumption of different types of active servers. At the model level, this allows servers to adjust power and computational performance through DVFS technology.

$$P_{\mathrm{sid}}^{tp} = c_{\mathrm{sid},1} u^{tp} \left(f^{tp}\right)^3 + c_{\mathrm{sid},2} = c_{\mathrm{sid},1} \frac{\lambda^{tp}}{\mu} \left(f^{tp}\right)^2 + c_{\mathrm{sid},2}, \tag{6}$$

where, superscript $tp \in \{\mathrm{a}, \mathrm{h}, \mathrm{b}\}$ identifies the types of active servers; specifically, a, h, and b respectively correspond to the active servers of Type A, Type H, and Type B. P_{sid}^{tp} represents the sum of idle and dynamic computing power of an active server, in W. $f^{tp} \in [0,1]$ is the relative value of the operating frequency of a specific type of active server, in dimensionless. When the actual operating frequency is 0, $f^{tp} = 0$; when the actual operating frequency is equal to the rated frequency of the chip, $f^{tp} = 1$. λ^{tp} represents the arrival rates of workload allocated to a specific type of active server, in unit/s. $u^{tp} = \frac{\lambda^{tp}}{\mu f^{tp}}$ is the utilization of a specific type of active server within a data center module, in dimensionless. $c_{\mathrm{sid},1} = 492$ W and $c_{\mathrm{sid},2} = 127$ W are model parameters [39].

Considering the impact of temperature-dependent leakage power, the computing power of different types of active servers within a data center module (P_{cp}^{tp}, in W) can be calculated by the following equation [48]:

$$P_{\mathrm{cp}}^{tp} = F_{\mathrm{T}}\left(T_{\mathrm{c}}^{tp}\right) P_{\mathrm{sid}}^{tp}, \tag{7}$$

where, T_{c}^{tp} represents the temperature of the chip within a specific type of active server, in °C; $F_{\mathrm{T}}(\cdot)$ is a function that describes impact of chip temperature on server power. As shown in Eq. (8), this study intends to adopt a linear model, which is reported to have an error of less than 1 % [49], to describe the aforementioned dependency relationship.

$$F_{\mathrm{T}}\left(T_{\mathrm{c}}^{tp}\right) = c_{\mathrm{lk},1} T_{\mathrm{c}}^{tp} + c_{\mathrm{lk},2}, \tag{8}$$

where, $c_{\mathrm{lk},1} = 3.125 \times 10^{-3}$ and $c_{\mathrm{lk},2} = 0.75$ are model constants [50].

Based on the fan affinity laws, the power of the fan array inside different types of active servers within a data center module (P_{sf}^{tp}, in W) can be described by a cubic function of its rotational speed (π^{tp}, in rpm), as shown in Eq. (9):

$$P_{\mathrm{sf}}^{tp} = P_{\mathrm{sf}}^{[\mathrm{r}]} \cdot \left(\frac{\pi^{tp}}{\pi^{[\mathrm{r}]}}\right)^3, \tag{9}$$

where, $\pi^{[\mathrm{r}]} = 9000$ rpm and $P_{\mathrm{sf}}^{[\mathrm{r}]} = 6 \times 5.64 = 33.84$ W represent the reference rotational speed and the reference power at that speed for the cooling fan array inside the server, respectively [36,38]. Taking into account the power of all types of servers, the power consumed by computing system of a data center module (P_{cps}, in W) can be calculated as follows:

$$P_{\mathrm{cps}} = \sum_{tp} N^{tp} \cdot P_{\mathrm{s}}^{tp} = \sum_{tp} N^{tp} \cdot \left(P_{\mathrm{cp}}^{tp} + P_{\mathrm{sf}}^{tp}\right), \tag{10}$$

where, N^{tp} indicates the number of active servers of different types within a data center module, in units; P_{s}^{tp} is the power of an active server within the module, in W.

3.2.3. Power loss in power conditioning units

The PDUs are central to ensuring that data centers operate continuously and efficiently. They take electricity from the public utility grid and deliver stable, reliable power to computing facilities within data centers through various conversion and distribution steps. Power losses by PDUs and UPS within a data center module (denoted as $P_{\mathrm{PDU,loss}}$ and $P_{\mathrm{UPS,loss}}$, in W, respectively), as key components of PDUs, can be calculated respectively by Eqs. (11) and (12) as follows [11,40]:

$$P_{\mathrm{PDU,loss}} = P_{\mathrm{PDU,idle}} + \eta_{\mathrm{PDU}}\left(\sum P_{\mathrm{s}}\right)^2, \tag{11}$$

$$P_{\mathrm{UPS,loss}} = P_{\mathrm{UPS,idle}} + \eta_{\mathrm{UPS}}\left(\sum P_{\mathrm{PDU}}\right), \tag{12}$$

where, $\eta_{\mathrm{PDU}} = \dfrac{1.5\ \%}{97\ \% \cdot \sum P_{\mathrm{s}}^{[\mathrm{r}]}}$ and $\eta_{\mathrm{UPS}} \approx 0.054945$ denote the power loss coefficients of a PDU and a UPS device, dimensionless, respectively; $\sum P_{\mathrm{s}}$ and $\sum P_{\mathrm{PDU}}$ represent the cumulative power of all servers and PDUs connected and powered by the PDU and UPS, in W, respectively; while $\sum P_{\mathrm{s}}^{[\mathrm{r}]}$ and $\sum P_{\mathrm{PDU}}^{[\mathrm{r}]}$ represent their rated values. $P_{\mathrm{PDU,idle}} = 1.5\ \% \cdot \dfrac{\sum P_{\mathrm{s}}^{[\mathrm{r}]}}{97\ \%}$ and $P_{\mathrm{UPS,idle}} = 4\ \% \cdot \dfrac{\sum P_{\mathrm{PDU}}^{[\mathrm{r}]}}{91\ \%}$ are the power consumption of PDU and UPS devices in standby mode, in W, respectively. Then, the total power consumed by the non-cooling system within a data center module (P_{ncl}, in W) can be calculated using the following equation:

$$P_{\mathrm{ncl}} = P_{\mathrm{cps}} + P_{\mathrm{PDU,loss}} + P_{\mathrm{UPS,loss}}. \tag{13}$$

3.2.4. Cooling power

The power consumed by CRAC unit within a data center module (P_{cl}, in W) can be calculated using the following equation:

$$P_{\mathrm{cl}} = \frac{P_{\mathrm{ncl}}}{CoP_{\mathrm{cl}}}, \tag{14}$$

where, CoP_{cl} represents the CoP of the CRAC in a data center module, which is dimensionless. It depends on the value of supply air temperature (T_{SA}, in °C). According to reference [41], it can be taken as:

$$CoP\left(T_{\mathrm{SA}}\right) = 0.0068 \cdot \left(T_{\mathrm{SA}}\right)^2 + 0.0008 \cdot T_{\mathrm{SA}} + 0.458. \tag{15}$$

3.2.5. Server thermal performance

Chip temperature threshold protection is critical for maintaining processor safety and stability. Elevated temperatures not only compromise the long-term reliability of processors but also lead to immediate server malfunctions, such as shutdowns [51]. Furthermore, chip temperature significantly influences leakage power, as detailed in Eqs. (7) and (8). Consequently, an accurate model of server thermal performance is essential for precise temperature estimation of the chip. Here, a model based on heat transfer theory [37,47], as shown in Eqs. (16) and (17), is used to estimate the chip temperature.

$$T_{\mathrm{c}}^{tp} = T_{\mathrm{SA}} + P_{\mathrm{c}}^{tp} R_{\mathrm{c\text{-}a}}^{tp}, \tag{16}$$

$$R_{\mathrm{c\text{-}a}}^{tp} = R_{\mathrm{c\text{-}h}}^{tp} + R_{\mathrm{h\text{-}a}}^{tp} = c_{\mathrm{hs},1} + \frac{c_{\mathrm{hs},2}}{\left(\dot{v}_{\mathrm{c}}^{tp}\right)^{c_{\mathrm{hs},3}}}, \tag{17}$$

where, P_{c}^{tp} is the power of the chip inside a specific type of active server, in W; $R_{\mathrm{c\text{-}a}}^{tp}$ represents the thermal resistance between the chip and the ambient (cooling) air inside different types of active servers, in °C/W; $R_{\mathrm{c\text{-}h}}^{tp}$ represents the thermal resistance between the chip and the heatsink, in °C/W; $R_{\mathrm{h\text{-}a}}^{tp}$ represents the convective heat transfer resistance between the heatsink and the ambient (cooling) air inside a specific type of active server, in °C/W. It depends on the volumetric flow rate of the cooling air passing through the chip ($\dot{v}_{\mathrm{c}}^{tp}$, in $\mathrm{m^3/s}$); $c_{\mathrm{hs},1} = 0.106$, $c_{\mathrm{hs},2} = 1.051 \times 10^{-3}$, and $c_{\mathrm{hs},3} = 0.938$ are parameters of the model, and their values are provided by the device manufacturer [37].

Given the assumption that the workload is uniformly distributed across the two chips within the server [36], and that the computing power of the server is directly proportional to the power of the chips [47], the power usage of an individual chip can be deduced using a linear equation as delineated in Eq. (18). Additionally, with the assumption of cooling air moving uniformly across the chips, the volumetric flow rate of air through a chip can be determined as indicated in Eq. (19):

$$P_{\mathrm{c}}^{tp} = \frac{P_{\mathrm{cp}}^{tp}}{P_{\mathrm{cp}}^{[\mathrm{r}]}} \cdot P_{\mathrm{c}}^{[\mathrm{TDP}]}, \tag{18}$$

$$\dot{v}_{\mathrm{c}}^{tp} = \frac{\dot{v}_{\mathrm{sf}}^{tp}}{2} = \frac{c_{\mathrm{sf}} \pi^{tp}}{2}, \tag{19}$$

where, $P_{\mathrm{c}}^{[\mathrm{TDP}]} = 270$ W denotes the thermal design power of a server chip [35,39]; $P_{\mathrm{cp}}^{[\mathrm{r}]} = 619$ W represents the rated computing power of a server [39]; $\dot{v}_{\mathrm{sf}}^{tp}$ represents the airflow rate of cooling air, which is driven by an array of fans and passes through a specific type of active server, in $\mathrm{m^3/s}$; $c_{\mathrm{sf}} = 1.029 \times 10^{-5}$ represents the parameter of a linear model that links the speed of the fan array to the air flow rate [38].

3.3. Formulation of the energy optimization problem

The optimization problem to minimize the energy consumption for data center module (P_{m}, in W) can be formulated as follows:

$$\begin{aligned} \min_{a^{\mathrm{a}}, b^{\mathrm{b}}, T_{\mathrm{SA}}, N^{tp}, f^{tp}, \pi^{tp}} P_{\mathrm{m}} &= P_{\mathrm{ncl}} + P_{\mathrm{cl}} \\ &= \left[P_{\mathrm{PDU,loss}} + P_{\mathrm{UPS,loss}} + \sum_{tp} N^{tp} \cdot \left(P_{\mathrm{cp}}^{tp} + P_{\mathrm{sf}}^{tp}\right)\right]\left(1 + \frac{1}{CoP\left(T_{\mathrm{cl}}\right)}\right). \end{aligned} \tag{20}$$

subject to:

$$0 \le a^{\mathrm{a}} \le a; \tag{20-a}$$

$$0 \le b^{\mathrm{b}} \le b; \tag{20-b}$$

$$T_{\mathrm{SA}}, \min \le T_{\mathrm{SA}} \le T_{\mathrm{SA}}, \max; \tag{20-c}$$

$$N^{tp} \in \{0, 1, \ldots, N_{\mathrm{m,s}}\},\ \forall tp \in \{\mathrm{a}, \mathrm{h}, \mathrm{b}\}; \tag{20-d}$$

$$\sum_{tp} N^{tp} \le N_{\mathrm{m,s}}; \tag{20-e}$$

$$0 \le f^{tp} \le f_{\max},\ \forall tp \in \{\mathrm{a}, \mathrm{h}, \mathrm{b}\}; \tag{20-f}$$

$$0 \le \pi^{tp} \le \pi_{\max},\ \forall tp \in \{\mathrm{a}, \mathrm{h}, \mathrm{b}\}; \tag{20-g}$$

$$u^{tp} \le 1,\ \forall tp \in \{\mathrm{a}, \mathrm{h}, \mathrm{b}\}; \tag{20-h}$$

$$t_{\mathrm{D}}^{tp} \le D,\ \forall tp \in \{\mathrm{a}, \mathrm{h}\}; \tag{20-i}$$

$$T_{\mathrm{c}}^{tp} \le T_{\mathrm{c}}, \max,\ \forall tp \in \{\mathrm{a}, \mathrm{h}, \mathrm{b}\}. \tag{20-j}$$

Inequalities (20-a) and (20-b) represent the upper and lower bound constraints for the quantity of real-time interactive workload allocated to Type A servers and the batch jobs allocated to Type B servers within a data center module, respectively. Constraint (20-c) represents the supply air temperature requirements for air-cooled data centers, where $T_{\mathrm{SA,min}} = 18$ °C and $T_{\mathrm{SA,max}} = 27$ °C are respectively the upper and lower limits of the supply air temperature for data center cooling systems recommended by the American Society of Heating, Refrigerating

and Air Conditioning Engineers (ASHRAE) [52]. Constraints (20-d) to (20-e) specify the upper and lower limit requirements for the number of activated servers of each type within a data center module, where $N_{\mathrm{m,s}} = 100$ represents the number of servers in a single data center module. Constraints (20-f) and (20-g) are the constraints on the chip operating frequency and fan speed for each type of server in a data center module, respectively; Constraints (20-h) and (20-i) ensure that the response times of each type of server meet QoS requirements, where $D = 0.05$ s is the upper limit of the service response time required by the QoS constraint. Inequality (20-j) reflects the control requirements for the chip temperature in each type of server, where $T_{\mathrm{c,max}} = 83°\mathrm{C}$ is the upper limit of chip temperature set according to the chip's allowable temperature.

The interactive workload arrival rates and batch task volumes in the data center are represented by a_{DC} and b_{DC}, in unit/s, respectively. And the total energy consumption is noted as P_{DC}, in W. Using the index $j \in \{1, 2, \ldots, N_{\mathrm{m}}\}$—where $N_{\mathrm{m}} = 20$ represents the total number of data center modules—variables corresponding to specific data center modules are identified. To ensure that modules in inactive states can quickly transition to active states to respond swiftly to increased computational workload, it is assumed that the PCUs within these inactive modules remain in standby, not fully shut down. This assumption is consistent with typical configurations in high-reliability data centers, such as those classified as Uptime Tier III or IV [34]. Then the issue of data center level-energy optimization is formulated as follows:

$$\min_{a_j, b_j, a_j^{\mathrm{a}}, b_j^{\mathrm{b}}, T_{\mathrm{SA},j}, N_j^{tp}, f_j^{tp}, \pi_j^{tp}} P_{\mathrm{DC}} = \sum_{j}^{N_{\mathrm{m}}} P_{\mathrm{m},j} = \sum_{j}^{N_{\mathrm{m}}} \left(P_{\mathrm{ncl},j} + P_{\mathrm{cl},j}\right)$$
$$= \sum_{j}^{N_{\mathrm{m}}} \left[P_{\mathrm{PDU,loss},j} + P_{\mathrm{UPS,loss},j} + \sum_{tp} N_j^{tp} \cdot \left(P_{\mathrm{cp},j}^{tp} + P_{\mathrm{sf},j}^{tp}\right)\right] \left(1 + \frac{1}{CoP\left(T_{\mathrm{cl},j}\right)}\right), \tag{21}$$

subject to:

$$\sum_{j=1}^{N_{\mathrm{m}}} a_j = a_{\mathrm{DC}}, \tag{21-a}$$

$$\sum_{j=1}^{N_{\mathrm{m}}} b_j = b_{\mathrm{DC}}, \tag{21-b}$$

$$0 \le a_j^{\mathrm{a}} \le a_j, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}; \tag{21-c}$$

$$0 \le b_j^{\mathrm{b}} \le b_j, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}; \tag{21-d}$$

$$T_{\mathrm{SA,min}} \le T_{\mathrm{SA},j} \le T_{\mathrm{SA,max}}, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}; \tag{21-e}$$

$$N_j^{tp} \in \{0, 1, \ldots, N_{\mathrm{m,s}}\}, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}, \ tp \in \{\mathrm{a, h, b}\}; \tag{21-f}$$

$$\sum_{tp} N_j^{tp} \le N_{\mathrm{m,s}}, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}; \tag{21-g}$$

$$0 \le f_j^{tp} \le f_{\max}, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}, \ tp \in \{\mathrm{a, h, b}\}; \tag{21-h}$$

$$0 \le \pi_j^{tp} \le \pi_{\max}, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}, \ tp \in \{\mathrm{a, h, b}\}; \tag{21-i}$$

$$u_j^{tp} \le 1, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}, \ tp \in \{\mathrm{a, h, b}\}; \tag{21-j}$$

$$t_{\mathrm{D},j}^{tp} \le D, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}, \ tp \in \{\mathrm{a, h}\}; \tag{21-k}$$

$$T_{\mathrm{c},j}^{tp} \le T_{\mathrm{c,max}}, \ \forall j \in \{1, 2, \ldots, \mathrm{N_m}\}, \ tp \in \{\mathrm{a, h, b}\}; \tag{21-l}$$

where, equation constraints (21-a) and (21-b) respectively represent the total computational workload constraints of the data center. Constraints (21-c) through (21-l) expand the module-level energy optimization constraints (20-a) through (20-j) to encompass the entire data center-level energy optimization.

4. Simplified load modeling for demand response

In the first part of this section, a detailed analysis is conducted on the solution to the energy optimization problem to determine the optimal energy characteristics of the data center. Then, it establishes a simplified data center-level load model for the data center, specifically for demand response, and examines the implications of simplification losses. Finally, a representative procedure is introduced, in which a group of data centers participates in power system demand response through a hybrid mechanism that combines both price-based and incentive-based approaches.

4.1. Optimal energy consumption characteristics of data centers

For a complex nonlinear system, after conducting optimization for a specific purpose, the resulting behavior may exhibit linear or weakly nonlinear characteristics. Although the system is inherently nonlinear, optimization effectively reduces fluctuations in its operating state, thereby diminishing the prominence of nonlinear characteristics [53]. This weak nonlinearity phenomenon is widely observed in fields such as energy optimization and process control. This section analyzes the solutions of the energy optimization problem based on nonlinear models in the foundational layer, aiming to determine whether such nonlinearity reduction exists and to identify the nature of weakly nonlinear optimal energy consumption characteristics in data centers. Specifically, these analyses are carried out at both the module- and data center-level.

4.1.1. Module-level characteristics

As shown in Fig. 2, the solution to the module-level data center energy optimization problem (20) reveals that, regardless of the computational workload, the data center module activates only Type H servers, while neither Type A nor Type B servers—both designed for specific workload types—are utilized. This indicates that, under the energy-optimal operating mode, both latency-sensitive interactive workloads and latency-tolerant batch jobs are executed in a hybrid deployment on servers of the same type. For the data center model employed in this study, this finding may be a natural consequence—since hybrid servers provide greater scheduling flexibility compared to dedicated servers, it is reasonable that efficiency is higher in hybrid servers when analyzed using queueing theory. However, it is important to emphasize that the real challenge of utilizing hybrid servers lies beyond the scope of queueing theory. Specifically, it involves designing a job scheduling strategy

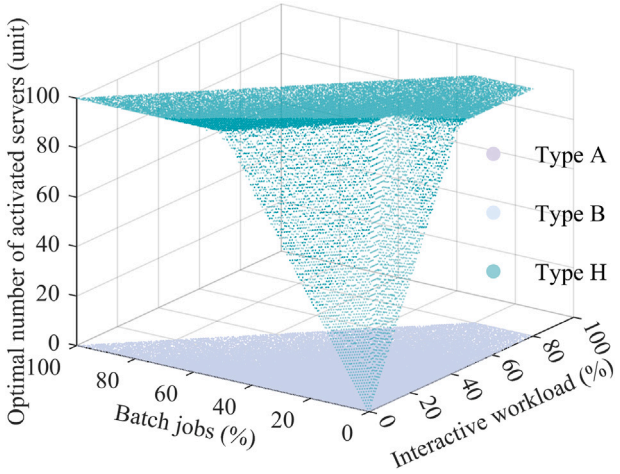


Fig. 2. Optimal activation patterns of servers under different computational workload conditions in a data center module. (Regardless of the computational workload conditions, the number of active Type A and Type B servers remains zero, thereby causing them to visually overlap in the figure.)

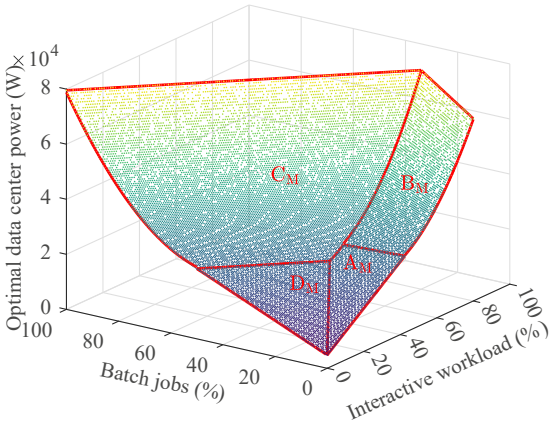


Fig. 3. Optimal power consumption characteristics of data center modules under varying computational workload conditions.

within the hybrid cluster to ensure that heterogeneous workloads can simultaneously satisfy their respective service constraints.

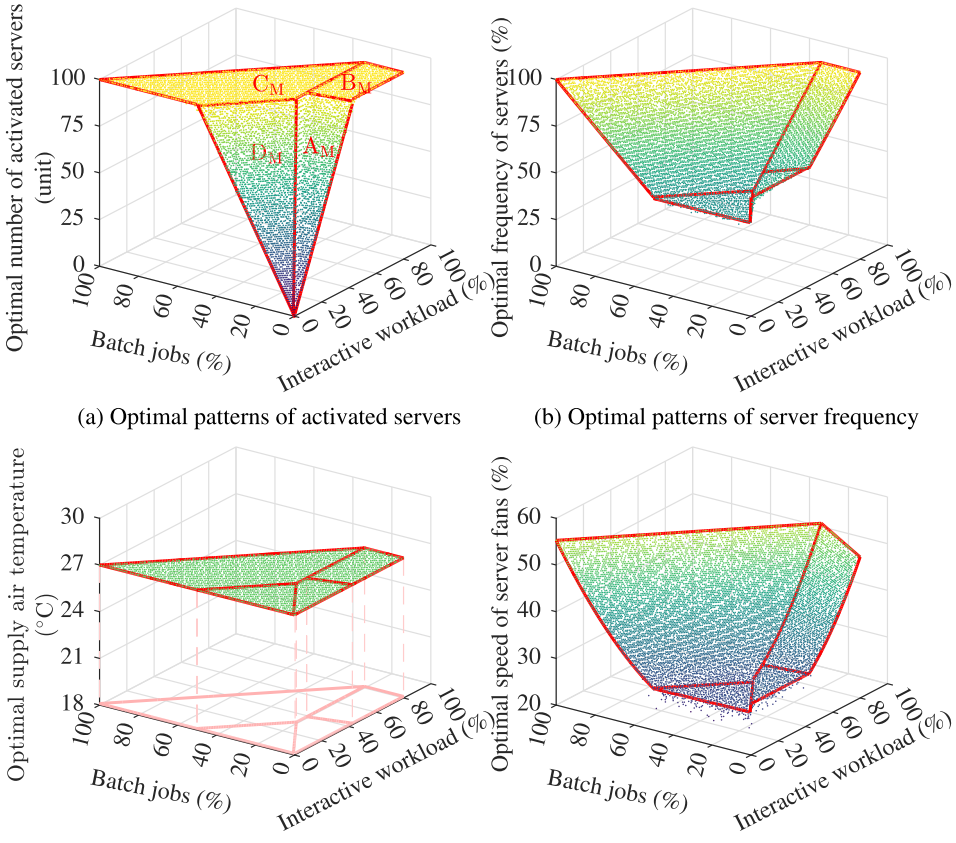
Figs. 3 and 4 depict how the optimal power consumption and other key operational parameters—such as the number of active Type H servers, their operating frequencies, fan speeds, and the supply air temperature of the cooling system—respond to changes in computational workload conditions. It can be observed from these figures that, the optimal energy consumption characteristics of the data center module can be categorized into four distinct modes: A_M, B_M, C_M, and D_M, depending on the computational workload conditions.

In region A_M, the optimal number of active servers increases linearly with rising interactive workload and batch jobs until it reaches the maximum limit of 100. The optimal operating frequency and fan speed of the activated servers display significant nonlinear characteristics with respect to computational workload conditions in this region. While the optimal power consumption of the data center module increases linearly with the growth of interactive workload and batch jobs in this area.

In region B_M, the optimal number of activated servers remains fixed at its maximum value. The optimal operating frequency of the servers increases linearly with the growth of interactive workload, while showing no dependency on the volume of batch jobs. The optimal fan speed of the servers exhibits a nonlinear (cubic) relationship with interactive workload and a linear relationship with batch jobs. Furthermore, the optimal power consumption of the data center module follows a pattern similar to the optimal fan speed of the servers under varying computational workload conditions.

In region C_M, the optimal operating frequency of servers exhibits a linear relationship, while the fan speed shows a nonlinear (cubic) relationship with the total computational workload, which includes both interactive tasks and batch jobs. While the number of activated servers still remains fixed at its maximum value. Furthermore, the optimal power consumption of the data center module increases nonlinearly, following a cubic trend, as the total computational workload grows.

In region D_M, resembling the conditions in region AM, the optimal number of activated servers increases linearly with the rising total computational workload until it reaches its maximum limit. The optimal operating frequency and fan speed of activated servers remain fixed and are unaffected by changes in computational demands. In this region, the



(a) Optimal patterns of activated servers

(b) Optimal patterns of server frequency

(c) Optimal patterns of supply air temperature

(d) Optimal patterns of server fan speed

Fig. 4. Optimal operational patterns of key decision variables under different computational workload conditions.

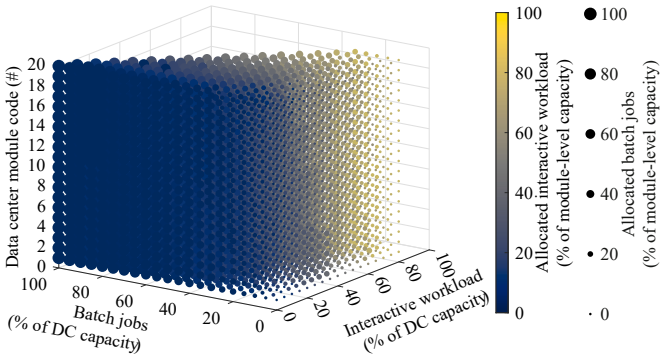


Fig. 5. Optimal workload distribution patterns among data center modules under different computational workload conditions.

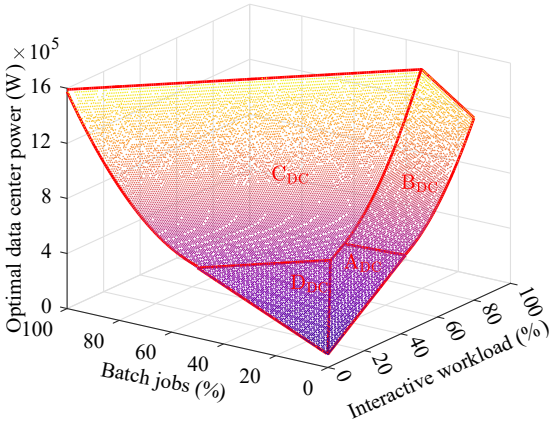


Fig. 6. Optimal power consumption characteristics of entire data centers under varying computational workload conditions.

optimal power consumption of the data center module grows linearly with the increase in total computational workload.

Notably, across all four regions, the optimal supply air temperature of the cooling system remains at 27 °C (the upper allowable limit) and is independent of computational workload conditions. Additionally, the cubic nonlinear relationships in region B_M and region C_M are considered to be dominated by Eq. (6).

4.1.2. Data center-level characteristics

Based on the solution to the optimization problem (21), Fig. 5 presents the optimal load allocation across data center modules under varying computational workload conditions. In this figure, the color of each scatter point indicates the proportion of interactive workload assigned to a specific module relative to its rated capacity, while the size of the point reflects the proportion of batch jobs relative to the rated capacity of the module. The figure reveals that under all computational workload conditions, the optimal allocation strategy tends to distribute various types of computational workloads evenly across all data center modules. This phenomenon is determined by the optimal load characteristics of the modules, where power consumption increases linearly with computational workload initially, regardless of load type, and transitions to cubic growth as the load continues to rise.

Fig. 6 depicts the optimal power consumption of the entire data center under various computational workload conditions. As shown in the figure, the optimal power consumption patterns of the data center can be categorized into four modes—A_{DC}, B_{DC}, C_{DC}, and D_{DC}—depending on computational workload conditions, resembling the patterns observed at the module level. Specifically, in region A_{DC}, the optimal power consumption grows linearly with the computational workload, whether interactive workload or batch jobs, but at distinct slopes. In region D_{DC}, the optimal power consumption increases linearly with the growing total computational workload. In region B_{DC}, the power consumption shows a nonlinear (cubic) relationship with interactive workload and a linear relationship with batch jobs. In region C_{DC}, the optimal power consumption exhibits a nonlinear (cubic) growth relationship with the total computational workload.

4.2. Simplified load modeling

Based on the observed optimal energy consumption characteristics at the data center level, the optimal power consumption under different computational workload conditions is fitted, resulting in a simplified load model for the data center. In particular, considering that cubic functions are difficult to solve and handle in optimization problems, this study uses quadratic functions to fit the nonlinear component of the load model, instead of the more physically interpretable cubic function. To demonstrate that the error introduced by this simplification is acceptable, the simplified load model is evaluated from two aspects: the coefficient of determination (R^2) and the relative error. Specifically, linear least-squares fitting is employed to establish the relationship between optimal power consumption and computational workload conditions in regions A_{DC} and D_{DC}, while nonlinear least-squares fitting (quadratic fitting) is used for regions B_{DC} and C_{DC}.

The resulting simplified load model, based on the optimal energy consumption characteristics, is expressed in Eq. (22). Table 3 outlines the details of the simplified load model for data centers, including the classification conditions for the four load patterns (A_{DC}, B_{DC}, C_{DC}, and D_{DC}), the identified model parameters, and the performance metrics. Particularly, the MREs of the simplified model across load patterns A_{DC}, B_{DC}, C_{DC}, and D_{DC} are 0.1404 %, 0.5368 %, 0.6259 %, and 0.0519 %, respectively. Additionally, the R^2 values in all four load patterns exceeded 0.9982, reflecting excellent model accuracy.

$$P_{DC}=\begin{cases} p_{A1}a_{DC}+p_{A2}b_{DC}+p_{A3}, & \text{for } (a_{DC},b_{DC})\in A_{DC} \\ p_{B1}{a_{DC}}^2+p_{B2}a_{DC}b_{DC}+p_{B3}a_{DC}+p_{B4}b_{DC}+p_{B5}, & \text{for } (a_{DC},b_{DC})\in B_{DC} \\ p_{C1}{c_{DC}}^2+p_{C2}c_{DC}+p_{C3}, & \text{for } (a_{DC},b_{DC})\in C_{DC} \\ p_{D1}c_{DC}+p_{D2}, & \text{for } (a_{DC},b_{DC})\in D_{DC} \end{cases}; \tag{22}$$

where, $c_{DC}=a_{DC}+b_{DC}$ is the total computational workload allocated to the data center; p_{A1}, p_{A2}, p_{A3}, p_{B1}, p_{B2}, p_{B3}, p_{B4}, p_{B5}, p_{C1}, p_{C2}, p_{C3}, p_{D1}, and p_{D2} are all the parameters of the simplified data center load model.

4.3. Demand response program

Fig. 7 depicts a typical scenario in which data centers participate in demand response. In this setting, data centers equipped with battery energy storage systems (BESSs) across multiple regions are connected to their respective regional power grids, which provide reliable power. Computational workload from various users, including latency-sensitive interactive loads and latency-tolerant batch jobs, is aggregated by front-end servers. The computational workload dispatch center manages and optimizes the distribution of these workloads across data centers, as well as the charge and discharge operations of their BESSs, based on TOU electricity pricing and demand response requests (associated with demand response events, DREs) from regional grids. This scenario integrates both price-based and incentive-based demand response mechanisms (TOU + interruptible/curtailable service or demand bidding/buy back).

For BESS in data center i, its state of charge (SoC) at time t (labeled as $SoC_{i,t}$, in dimensionless), can be calculated as:

$$SoC_{i,t}=SoC_{i,0}+\frac{\Delta t}{ES^{[r]}}\left(\eta_{ch}\sum_{r=1}^{t}P_{ch,i,r}-\frac{\sum_{r=1}^{t}P_{dis,i,r}}{\eta_{dis}}\right), \tag{23}$$

Table 3
Details of the simplified load model of the data center based on its optimal energy consumption characteristics.

Region	Conditions	Parameters	Feature	R^2	MRE [%]
A_{DC}	$Db_{DC} \le N_m N_{m,s}$, $b_{DC}(f_D \mu D - 1) \le a_{DC}^*$, and $b_{DC} + 4a_{DC} \le 1.7 N_m N_{m,s} \mu$	$p_{A1} = 5.4090$, $p_{A2} = 2.0737$, $p_{A3} = 9.5268 \times 10^4$.	Linear	0.9999	0.1404
B_{DC}	$Db_{DC} \le N_m N_{m,s}$, $Da_{DC} \le N_m N_{m,s}(\mu D - 1)$, and $b_{DC} + 4a_{DC} \ge 1.7 N_m N_{m,s} \mu$	$p_{B1} = 8.1508 \times 10^{-5}$, $p_{B2} = 5.5958 \times 10^{-5}$, $p_{B3} = -9.6214$, $p_{B4} = -2.7381$, $p_{B5} = 7.9076 \times 10^5$.	Quadratic, and non-convex	0.9982	0.5368
C_{DC}	$Db_{DC} \ge N_m N_{m,s}$, $a_{DC} + b_{DC} \le N_m N_{m,s} \mu$, and $a_{DC} + b_{DC} \ge N_m N_{m,s} f_D \mu^*$.	$p_{C1} = 7.9139 \times 10^{-5}$, $p_{C2} = -13.1268$, $p_{C3} = 1.0441 \times 10^6$.	Quadratic, and convex	0.9990	0.6259
D_{DC}	$b_{DC}(f_D \mu D - 1) \ge a_{DC}^*$, and $a_{DC} + b_{DC} \le N_m N_{m,s} f_D \mu^*$.	$p_{D1} = 4.1381$, $p_{D2} = 9.5268 \times 10^4$.	Linear	1.0000	0.0519

* Based on the module-level optimal energy efficiency characteristics of the data center, $f_D = 0.5$.

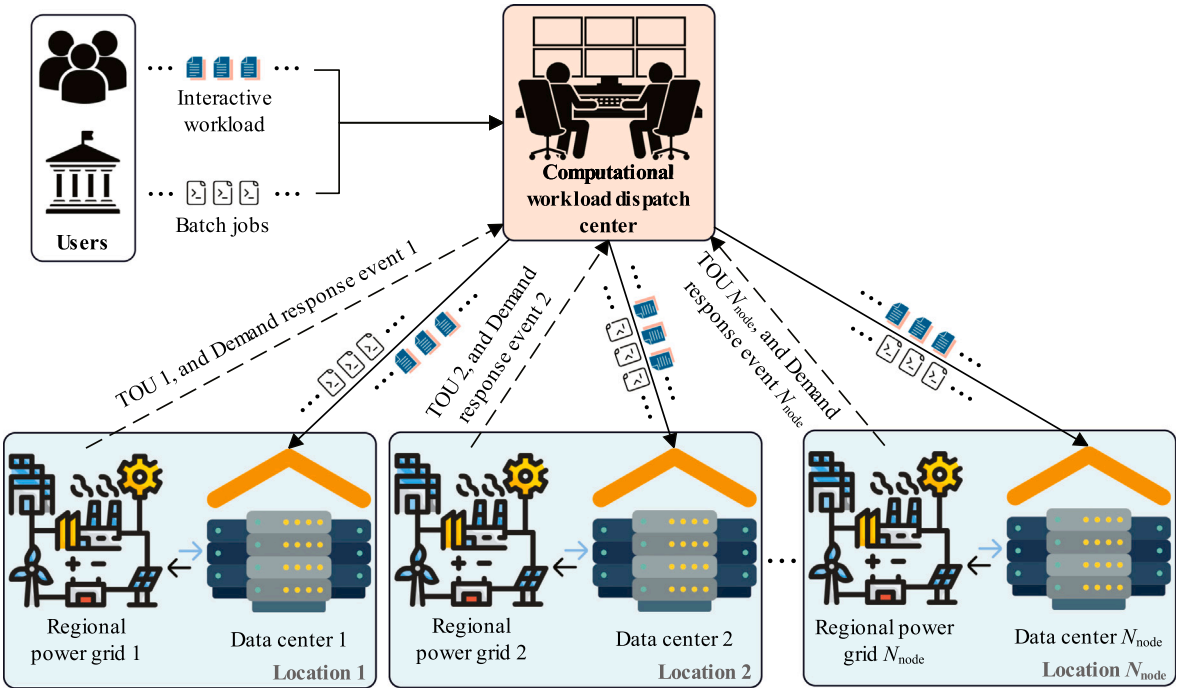


Fig. 7. Schematic diagram of scenarios where data centers participate in demand response.

where, subscript $i \in \{1, 2, \ldots, N_{node}\}$ represents the identifier of each data center, while $t \in \{1, 2, \ldots, 24\}$ denotes the optimization time slot corresponding to the hour on the event day; $SoC_{i,0} = 0$ is the initial SoC of the BESS in data center i; $\Delta t = 1$ h indicates the width of the optimization time slot; $ES^{[r]}$ is the rated capacity of the BESS, in Wh. Assuming that the BESS in each UPS can maintain full-load operation of its respective data center module for 10 min during a power outage [54], it follows that $ES^{[r]} = 2 \times (20 + 1) \times 100 \times \frac{10}{60} = 700$ kW. $P_{ch,i,r}$ and $P_{dis,i,r}$ represent the charging and discharging power of the BESS in data center i at time r, in W, respectively; $\eta_{ch} = 0.98$ and $\eta_{dis} = 0.96$ are the charging and discharging loss coefficients of the BESS, in dimensionless, respectively [55].

Then, the day-ahead scheduling problem for data centers engaging in power system demand response through a hybrid mechanism—combining price-based methods (TOU) and incentive-based approaches (e.g., participating in curtailable service or demand bidding, and establishing a contract with the grid to curtail or augment power loads during designated periods on the event day), as illustrated in Fig. 7—can be formulated as the following MINLP problem:

$$\min_{a_{i,t}, b_{i,t}, P_{ch,i,t}, P_{dis,i,t}} \sum_{i=1}^{N_{node}} c_{base,i} TC_i \Delta T + \sum_{i=1}^{N_{node}} \sum_{t=1}^{24} c_{i,t} \left(P_{DC,i,t}^{simp} + P_{ch,i,t} - P_{dis,i,t} \right) \Delta t, \tag{24}$$

subject to:

$$\sum_{i=1}^{N_{node}} a_{i,t} = a_{DCO,t}, \ \forall\, t \in \{1, 2, \ldots, 24\}; \tag{24-a}$$

$$\sum_{i=1}^{N_{node}} \sum_{t=1}^{24} b_{i,t} = b_{DCO}; \tag{24-b}$$

$$0 \le a_{i,t} \le \frac{(\mu D - 1) N_m N_{m,s}}{D}, \ \forall\, i \in \{1, 2, \ldots, N_{node}\}, \ t \in \{1, 2, \ldots, 24\}; \tag{24-c}$$

$$0 \le b_{i,t} \le N_m N_{m,s} \mu - a_{i,t}, \ \forall\, i \in \{1, 2, \ldots, N_{node}\}, \ t \in \{1, 2, \ldots, 24\}; \tag{24-d}$$

$$0 \le P_{\mathrm{ch},i,t} \le P_{\mathrm{ch,max}},\ \forall\, i \in \{1,2,\dots,N_{\mathrm{node}}\},\ t \in \{1,2,\dots,24\}; \tag{24-e}$$

$$0 \le P_{\mathrm{dis},i,t} \le P_{\mathrm{dis,max}},\ \forall\, i \in \{1,2,\dots,N_{\mathrm{node}}\},\ t \in \{1,2,\dots,24\}; \tag{24-f}$$

$$P_{\mathrm{ch},i,t} \cdot P_{\mathrm{dis},i,t} = 0,\ \forall\, i \in \{1,2,\dots,N_{\mathrm{node}}\},\ t \in \{1,2,\dots,24\}; \tag{24-g}$$

$$P^{\mathrm{simp}}_{\mathrm{DC},i,t} + P_{\mathrm{ch},i,t} - P_{\mathrm{dis},i,t} \ge 0,\ \forall\, i \in \{1,2,\dots,N_{\mathrm{node}}\},\ t \in \{1,2,\dots,24\}; \tag{24-h}$$

$$0 \le SoC_{i,t} \le 1,\ \forall\, i \in \{1,2,\dots,N_{\mathrm{node}}\},\ t \in \{1,2,\dots,24\}; \tag{24-i}$$

$$SoC_{i,24} = SoC_{i,0},\ \forall\, i \in \{1,2,\dots,N_{\mathrm{node}}\}; \tag{24-j}$$

$$\sum_{t=t_{\mathrm{start},i}}^{t_{\mathrm{end},i}} \left[P^{*}_{\mathrm{DCES},i,t} - \left(P^{\mathrm{simp}}_{\mathrm{DC},i,t} + P_{\mathrm{ch},i,t} - P_{\mathrm{dis},i,t}\right)\right] \cdot \Delta t = CAE_i,\ \forall\, i \in I_{\mathrm{DRE}}. \tag{24-k}$$

where, N_{node} denotes the number of data centers managed by the data center operator (computational workload dispatch center), in unit; $a_{i,t}$ and $b_{i,t}$ represent the arrival rate of interactive workload and the volume of batch jobs allocated to data center i at time t, in unit/s, respectively. $c_{\mathrm{base},i}$ and $c_{i,t}$ are the base and TOU electricity rates for data center i at time t, in Yuan/(kVA·month) and Yuan/Wh, respectively. TC_i represents the transformer capacity of data center i, in kVA; $P^{\mathrm{simp}}_{\mathrm{DC},i,t} = P^{\mathrm{simp}}_{\mathrm{DC}}(a_{i,t}, b_{i,t})$ is the power of data center i at time t, calculated based on the data center load model (i.e., Eqs. 22 or A8 in this study), in W. $\Delta T = 1\ 30$ month denotes equivalent amount of time in a day relative to a month.

Constraints (24-a) and (24-b) define the total limits for interactive workload and batch jobs, respectively. $a_{\mathrm{DCO},t}$ indicates the predicted arrival rate of interactive workload received by the data center operator at time t on the event day, in unit/s. b_{DCO} represents the total volume of batch jobs that the data center operator receives before the event day and needs to complete on that day, in unit. Constraints (24-c) and (24-d) specify the capacity constraints of data center i at time t for handling interactive workload and batch jobs, which depends on the QoS constraints. Inequalities (24-e) and (24-f) specify the upper and lower bounds for the charging and discharging power of the BESS in data center i at time t. $P_{\mathrm{ch,max}}$ and $P_{\mathrm{dis,max}}$ denote the maximum charging and discharging power limits for the BESS, in W, respectively. Eq. (24-g) sets the charge-discharge state constraint, ensuring that charging and discharging do not occur simultaneously. Given the limited capacity of the BESS integrated within the data center and the critical need to maintain a high SoC for operational safety, the reverse sale of power from the BESS to the grid is excluded, as specified by constraint (24-h). Constraints (24-i) and (24-j) define the SoC requirements, with Eq. (24-j) specifically requiring the BESS to return to its initial SoC at the end of the event day. Constraint (24-k) specify the event time and contracted curtailment or augmentation of electricity for the data center during the DRE. $t_{\mathrm{start},i}$ and $t_{\mathrm{end},i}$ denote the start and end times for the participation of data center i in the DRE, respectively. I_{DRE} represents the set of indices of data centers participating in the DRE. CAE_i refers to the contracted curtailment or augmentation of electricity of data center i during the DRE period, with positive values indicating curtailment and negative values indicating augmentation, in Wh. $P^{*}_{\mathrm{DCES},i,t}$ is the customer baseline (power) load data center i and its BESS at time t, in W, calculated as follows:

$$P^{*}_{\mathrm{DCES},i,t} = P^{\mathrm{simp}}_{\mathrm{DC}}\left(a^{*}_{i,t}, b^{*}_{i,t}\right) + P^{*}_{\mathrm{ch},i,t} - P^{*}_{\mathrm{dis},i,t}, \tag{25}$$

where,

$$\begin{aligned}&\left(a^{*}_{i,t}, b^{*}_{i,t}, P^{*}_{\mathrm{ch},i,t}, P^{*}_{\mathrm{dis},i,t}\right)\\ &= \arg \min_{a_{i,t},b_{i,t},P_{\mathrm{ch},i,t},P_{\mathrm{dis},i,t}} \sum_{i=1}^{N_{\mathrm{node}}} c_{\mathrm{base},i} TC_i \Delta T\\ &\quad + \sum_{i=1}^{N_{\mathrm{node}}} \sum_{t=1}^{24} c_{i,t}\left(P^{\mathrm{simp}}_{\mathrm{DC},i,t} + P_{\mathrm{ch},i,t} - P_{\mathrm{dis},i,t}\right)\Delta t,\end{aligned} \tag{25-a}$$

subject to: Constraints (24-a) to (24-j).

Eq. (25) and its associated optimization problem (25-a) define the baseline load as the solution to a cost minimization problem that considers both the base electricity rate and the time-varying TOU price, aiming to reflect the realistic operating behavior of the data center under typical electricity pricing schemes.

Considering the non-convex nature of the bilinear term, when the demand response programming problem (24) adopts the simplified load model based on the optimal energy consumption characteristics of the data center proposed in this study (as shown in Eq. (22)), the piecewise McCormick relaxation method [57,58] with K-pieces is used to convexly approximate the bilinear term "$p_{\mathrm{B2}} a_{\mathrm{DC}} b_{\mathrm{DC}}$" in the model, as follows:

$$p_{\mathrm{B2}} a_{\mathrm{DC}} b_{\mathrm{DC}} = \begin{cases} p_{\mathrm{B2}} w^{1}_{\mathrm{DC}}, & \text{for } a_{\mathrm{DC}} \in \left[a^{\mathrm{B},1}_{\mathrm{DC,min}}, a^{\mathrm{B},1}_{\mathrm{DC,max}}\right] \text{ and } b_{\mathrm{DC}} \in \left[b^{\mathrm{B},1}_{\mathrm{DC,min}}, b^{\mathrm{B},1}_{\mathrm{DC,max}}\right] \\ p_{\mathrm{B2}} w^{2}_{\mathrm{DC}}, & \text{for } a_{\mathrm{DC}} \in \left[a^{\mathrm{B},2}_{\mathrm{DC,min}}, a^{\mathrm{B},2}_{\mathrm{DC,max}}\right] \text{ and } b_{\mathrm{DC}} \in \left[b^{\mathrm{B},2}_{\mathrm{DC,min}}, b^{\mathrm{B},2}_{\mathrm{DC,max}}\right] \\ \vdots & \\ p_{\mathrm{B2}} w^{k}_{\mathrm{DC}}, & \text{for } a_{\mathrm{DC}} \in \left[a^{\mathrm{B},k}_{\mathrm{DC,min}}, a^{\mathrm{B},k}_{\mathrm{DC,max}}\right] \text{ and } b_{\mathrm{DC}} \in \left[b^{\mathrm{B},k}_{\mathrm{DC,min}}, b^{\mathrm{B},k}_{\mathrm{DC,max}}\right] \\ \vdots & \\ p_{\mathrm{B2}} w^{K}_{\mathrm{DC}}, & \text{for } a_{\mathrm{DC}} \in \left[a^{\mathrm{B},K}_{\mathrm{DC,min}}, a^{\mathrm{B},K}_{\mathrm{DC,max}}\right] \text{ and } b_{\mathrm{DC}} \in \left[b^{\mathrm{B},K}_{\mathrm{DC,min}}, b^{\mathrm{B},K}_{\mathrm{DC,max}}\right] \end{cases}; \tag{26}$$

where,

$$\begin{cases} w^{k}_{\mathrm{DC}} \ge a^{\mathrm{B},k}_{\mathrm{DC,min}} b_{\mathrm{DC}} + a_{\mathrm{DC}} b^{\mathrm{B},k}_{\mathrm{DC,min}} - a^{\mathrm{B},k}_{\mathrm{DC,min}} b^{\mathrm{B},k}_{\mathrm{DC,min}} \\ w^{k}_{\mathrm{DC}} \ge a^{\mathrm{B},k}_{\mathrm{DC,max}} b_{\mathrm{DC}} + a_{\mathrm{DC}} b^{\mathrm{B},k}_{\mathrm{DC,max}} - a^{\mathrm{B},k}_{\mathrm{DC,max}} b^{\mathrm{B},k}_{\mathrm{DC,max}} \\ w^{k}_{\mathrm{DC}} \le a^{\mathrm{B},k}_{\mathrm{DC,max}} b_{\mathrm{DC}} + a_{\mathrm{DC}} b^{\mathrm{B},k}_{\mathrm{DC,min}} - a^{\mathrm{B},k}_{\mathrm{DC,max}} b^{\mathrm{B},k}_{\mathrm{DC,min}} \\ w^{k}_{\mathrm{DC}} \le a^{\mathrm{B},k}_{\mathrm{DC,min}} b_{\mathrm{DC}} + a_{\mathrm{DC}} b^{\mathrm{B},k}_{\mathrm{DC,max}} - a^{\mathrm{B},k}_{\mathrm{DC,min}} b^{\mathrm{B},k}_{\mathrm{DC,max}} \end{cases}, \quad \forall\, k \in \{1,2,\dots,K\}. \tag{26-a}$$

5. Simulation

In this section, the proposed weakly nonlinear load model based on optimal energy consumption characteristics, along with the conventional linear model, is embedded into the demand response program to verify the effectiveness of the proposed modeling method.

5.1. Simulation setups

This section outlines the simulation setups in detail. Specifically, it includes the setup of a demand response scenario with two data centers and the setup of the comparative experiment.

5.1.1. Demand response scenario setups

This simulation experiment considers two hypothetical data centers equipped with BESSs located in Beijing and Zhangjiakou, one of the cities designated for data center clusters under the "Eastern Data and Western Computing" initiative of China, with system configurations as described in Section 3.1 of this work. The two-part TOU electricity pricing of the respective regional power grids is presented in Table 4, based on the electricity price list for industrial and commercial users (220 kV and above) released by the power company in July 2024 for August. The data center located in Beijing is assumed to have reached a demand response contract with the grid by participating in either a curtailable service or a demand bidding program, requiring load curtailment during specific time periods on the event day. As shown in Table 4, it is assumed that the event day is August 1, 2024, the hottest day of the month. The event time and the contracted curtailment of electricity are set at 16:00 to 21:00 and 2000 kWh, respectively, to mitigate the grid load during the summer peak period.

In addition, as illustrated in Fig. 8, this study utilizes normalized and scaled traffic intensity data from the Google Hong Kong search service [56] on the event day to represent the predicted arrival rate of interactive workloads processed by the data center operator. Meanwhile, the total volume of batch jobs that need to be processed on the event day is set to 30 % of the rated computational capacity of the operator.

Table 4
Demand response environment settings in simulation experiments.

Data center (#)	Location	Transformer capacity (kVA)	Event day and time	Contracted curtailment (kWh)	Basic electricity price (Yuan/ (kVA · month))	Time period (h)	Electricity price (Yuan/kWh)
1	Beijing	2000	August 1, 2024, 16:00–21:00	2000	28	00:00–07:00, 23:00–24:00	0.3514
						07:00–10:00, 13:00–16:00, 22:00–23:00	0.5943
						10:00–11:00, 17:00–22:00	0.8372
						11:00–13:00, 16:00–17:00	0.9668
2	Zhangjiakou	2000	–	–	21.6	00:00–07:00, 23:00–24:00	0.2711
						07:00–10:00, 12:00–16:00, 22:00–23:00	0.5639
						10:00–12:00, 16:00–17:00, 20:00–22:00	0.8567
						17:00–20:00	0.9989

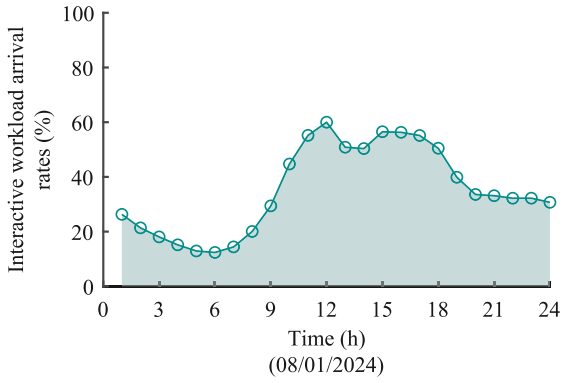


Fig. 8. Interactive workload arrival rates on the event day.

5.1.2. Comparative experimental setups

To validate the effectiveness of the load model proposed in this study, both the simplified load model, which is based on the optimal energy consumption characteristics of data centers, and the conventional linear load model are integrated into the demand response program outlined in Section 4.3. The problem is then solved using commercial MINLP solvers such as Gurobi [59] to derive dispatch decisions that allocate various computational workloads over time and across different spatial locations (i.e., data centers situated at different locations). Subsequently, the workloads associated with these dispatch decisions are input into mathematical models that represent the specific operational modes of the data centers across different scenarios, to assess their performance in demand response program.

Table 5 presents the scenario settings for the comparative experiments. Scenario 1 aims to assess the performance of a data center in a demand response environment when the conventional linear load model is used for programming, and the data center operates in a mode that aligns with the energy consumption characteristics predicted by the model. While this operating mode is energy-inefficient, it theoretically guarantees the response quality of the data center during the DRE when the conventional linear load model is used for demand response programming. Scenario 2 evaluates the performance of the data center when the conventional linear load model is used in demand response programming, and the data center operates in the optimal energy-efficient mode. Scenario 3 evaluates the performance of the data center when the simplified nonlinear load model, proposed in this study and based on the optimal energy consumption characteristics of data centers, is employed for demand response programming, with the data center operating in the optimal energy-efficient mode.

5.2. Results and discussion

This section begins by introducing demand response strategies based on different models. It then provides a detailed discussion and analysis of the performance of data centers in demand response across various scenarios, with a focus on electricity costs, energy consumption, and demand response quality.

5.2.1. Demand response strategy

Fig. 9 presents the dispatching decisions/response strategies in the demand response program with different data center load models. This figure also highlights the influence of DREs (as detailed in Table 4) on these strategies. Specifically, Fig. 9(a) and (b) depict the allocation of computational workload between Data Center 1 (in Beijing) and Data Center 2 (in Zhangjiakou), alongside the charging and discharging operations of the energy storage system, under the conventional linear load model in demand response programming. Meanwhile, Fig. 9(c) and Fig. 9(d) illustrate the workload distribution and energy storage operations when the simplified nonlinear load model proposed in this study, derived from the optimal energy consumption characteristics of data centers, is utilized.

From the comparison between Fig. 9(a) and (b), as well as Fig. 9(c) and (d), it can be observed that Data Center 2 consistently processes more computational workload than Data Center 1, regardless of the occurrence of DREs. This disparity is primarily attributed to the TOU electricity pricing in Zhangjiakou, where Data Center 2 is located, which is generally lower than that in Beijing, the location of Data Center 1, for the majority of the day. The energy storage systems in the data centers at both locations undergo two charge-discharge cycles, a pattern closely aligned with the periodicity of the TOU electricity pricing. From the comparison between Fig. 9(a) and (c), as well as Fig. 9(b) and (d), it becomes evident that, compared to the conventional linear data center load model, the simplified nonlinear load model proposed in this

Table 5
Scenario setups in comparative experiments.

Model	Scenario		
	Scenario 1	Scenario 2	Scenario 3
Load model used in demand response program	Conventional linear model[a]	Conventional linear model[a]	Proposed simplified nonlinear model[b]
Data center operation mode in performance evaluation	According to conventional linear model[a]	Optimal energy-efficient mode[c]	Optimal energy-efficient mode[c]

[a] Refers to the model detailed in Appendix A.
[b] Refers to the model presented in Eq. (22) in Section 4.2.
[c] Refers to the operation of the data center after receiving the computational workload, as determined by the solution to the energy consumption optimization problem (21).

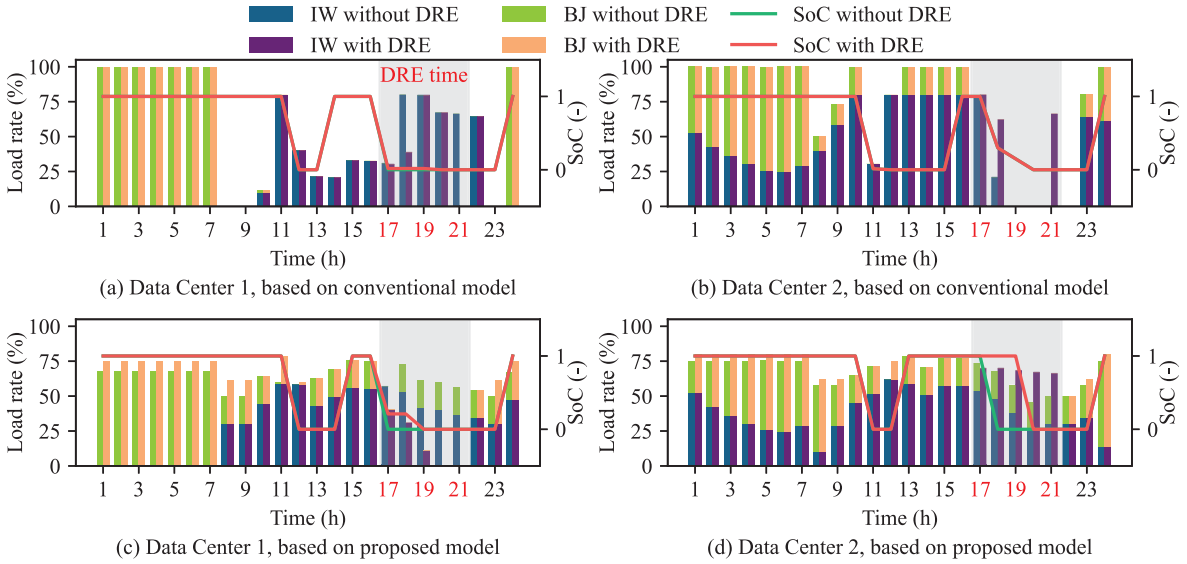


Fig. 9. Data center response strategies with and without DREs. (In the legend, the abbreviations IW and BJ represent interactive workloads and batch jobs, respectively.)

study prompts the computational workload dispatch center to avoid assigning excessive computational workload (exceeding about 80 % of the rated computational capacity) to any single data center in demand response program. Furthermore, when the conventional linear data center load model is used in demand response programming, the computational workload dispatch center achieves the contracted curtailment of electricity in Data Center 1 during the DRE time (16:00–21:00) solely by reallocating a portion of the interactive workload originally processed by Data Center 1 to Data Center 2. When employing the simplified nonlinear load model introduced in this work for demand response programming, to fulfill the contracted curtailment of electricity in Data Center 1 during the DRE, the workload dispatch center must not only transfer a portion of the interactive load originally handled by Data Center 1 to Data Center 2 but also shift batch jobs originally performed by Data Center 1 to other time periods within Data Center 1 and to Data Center 2 for execution.

5.2.2. *Electricity costs*

As shown in Fig. 10, the hourly and cumulative electricity costs of the two data centers under three experimental scenarios (as described in Table 5) are presented. It can be observed that, for the majority of the day (21 out of 24 hours), the total electricity costs of the two data centers in Scenario 1 exceed those in Scenarios 2 and 3. For 15 hours of the 24-hour day, the total electricity costs of the two data centers in Scenario 3 are lower than those in Scenarios 1 and 2. Specifically, during the DRE (16:00–21:00), the total electricity costs for both data centers in Scenario

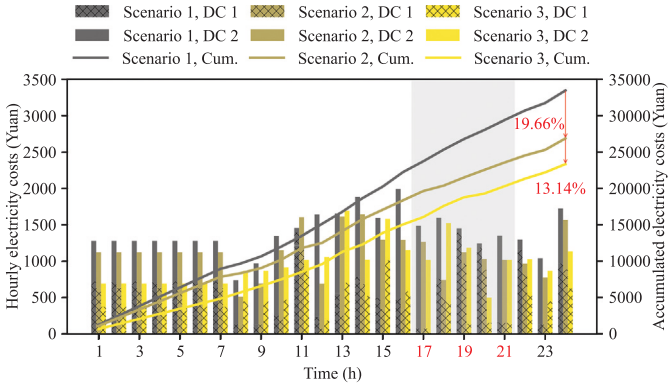


Fig. 10. Electricity costs across different data centers under different scenarios. (In the legend, the abbreviations Cum., and DC represent cumulative, and data center, respectively.)

1 remain the highest at all time points. The electricity costs in Scenario 2 and Scenario 3 fluctuate, with the cumulative costs during the DRE being nearly identical (5191.88 Yuan and 5256.24 Yuan, respectively). From a holistic perspective, compared to the daily cumulative electricity costs of the two data centers in Scenario 1, the costs in Scenario 2 decreased by 19.66 %, and the costs in Scenario 3 further declined by 13.14 % relative

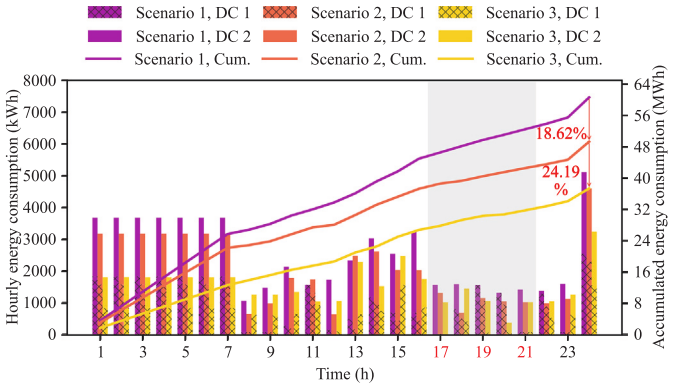


Fig. 11. Energy consumption across different data centers under different scenarios. (In the legend, the abbreviations Cum., and DC represent cumulative, and data center, respectively.)

to Scenario 2 (the cumulative electricity costs in Scenario 1, Scenario 2, and Scenario 3 are 33470.28 Yuan, 26890.15 Yuan, and 23357.49 Yuan, respectively).

5.2.3. Energy consumption

Fig. 11 presents the energy consumption underlying the hourly and cumulative electricity costs of each data center across the three experimental scenarios. Similar to the electricity costs discussed in the previous section, the total energy consumption of the two data centers in Scenario 1 is the highest for most of the day (21 out of 24 hours), compared to Scenarios 2 and 3. Scenario 3 records the lowest total energy consumption for the two data centers for the majority of the time (17 out of 24 hours). Regardless of the scenario, compared to the energy consumption in other time periods, the total energy consumption of the two data centers during the DRE is at a lower level. Specifically, in all three scenarios, the peak total energy consumption of the two data centers is observed between 23:00 and 24:00. The reason for this phenomenon is that, regardless of whether it is Beijing, where Data Center 1 is located, or Zhangjiakou, where Data Center 2 is located, the electricity price during this period is at its lowest (valley), while the previous period has a flat electricity rate. As a result, a larger portion of the computational workload is allocated to the two data centers, and the energy storage systems are charged during this time to restore their rated capacity by 24:00. From a daily cumulative perspective, the cumulative energy consumption of the two data centers in Scenario 2 is reduced by 18.62 % compared to Scenario 1 (from 60655.82 kWh to 49361.65 kWh), and the cumulative energy consumption in Scenario 3 is further reduced by 24.19 % compared to Scenario 2 (from 49361.65 kWh to 37420.68 kWh).

5.2.4. Demand response quality

The electricity curtailment deficit (ECD) is defined as the gap between the actual and desirable electricity curtailment of a data center during the DRE. This metric is used to assess the quality of demand response in data centers and is derived from the concept of power demand response deficit [20,60]. As shown in Eq. (27), according to the definition of the ECD, a smaller value indicates better demand response quality in the data center, whereas a larger value reflects poorer demand response quality.

$$ECD_t = \left| EC_t - \hat{EC}_t \right|, \tag{27}$$

where, ECD_t denotes the ECD in time slot t, while EC_t and $\hat{EC}_t$ represent the actual and target electricity curtailment of the data center during time slot t, respectively. The target value is derived from the results of the demand response programming.

Fig. 12 presents the demand response quality of Data Center 1 during the DRE across various time slots. In Fig. 12(a), the target and actual electricity curtailment, along with the gap between them, are displayed for different scenarios at each time point. Fig. 12(b) shows the ECDs and their cumulative values at each time point across different scenarios. The numerical details of these results are provided in Table 6. Fig. 12(a) reveals that, in Scenario 2 (which uses a conventional linear load model for demand response programming), there is a significant gap between the target and actual electricity curtailment across several time slots, particularly during the periods from 18:00 to 19:00 and from 20:00 to 21:00. In contrast, in Scenario 3 (which employs the simplified nonlinear load model proposed in this study, based on the optimal energy consumption characteristics of data centers), the electricity curtailment achieved by the data center in all time slots during the DRE closely matches the target values. This outcome is a direct consequence of the proposed load model, which more realistically reflects the characteristics of data center energy consumption.

Table 6 shows that during the entire DRE period (16:00–21:00, with a target electricity curtailment of 2000 kWh), the actual electricity curtailment achieved by the data center in Scenario 2 and Scenario 3 is 1906.13 kWh and 2013.17 kWh, respectively, with corresponding ECD

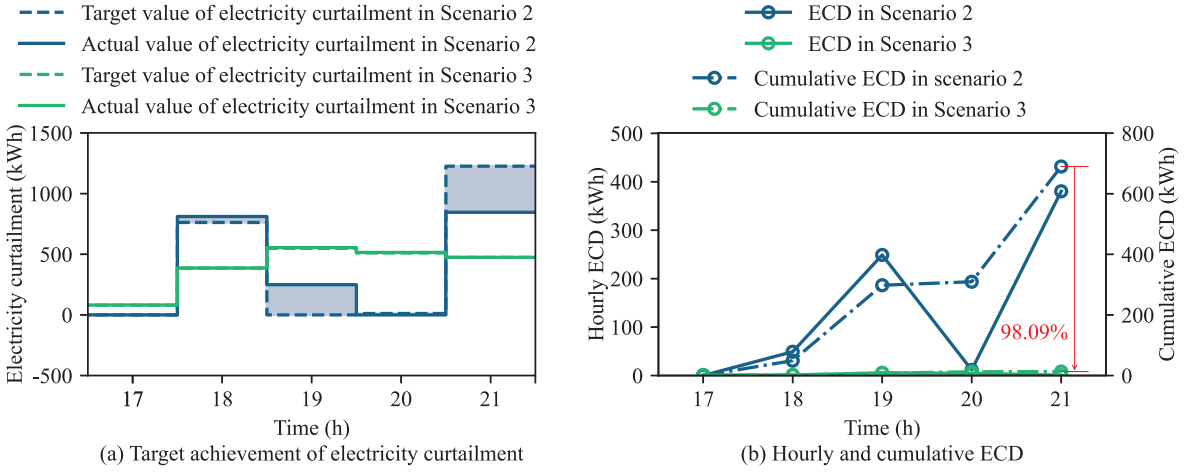


Fig. 12. Demand response quality of data centers during DREs under different scenarios. (In Scenario 1, because the conventional linear load model is used in both the demand response programming and operational simulation experiments, the target and actual electricity curtailment at each time point are identical. Consequently, the ECD equals zero and provides no comparative value, so it is not shown in this figure.)

Table 6
Details of the data center demand response quality in different scenarios.

Scenario	Time	Electricity curtailment (kWh)			
		Target	Actual	ECD	Cumulative ECD
Scenario 2	16:00–21:00 (Whole event time)	2000.00	1906.13	93.87	–
	16:00–17:00	0.00	0.00	0.00	0.00
	17:00–18:00	762.16	811.29	49.13	49.13
	18:00–19:00	0.00	248.90	248.90	298.03
	19:00–20:00	11.55	0.00	11.55	309.58
	20:00–21:00	1226.29	845.94	380.35	689.93
Scenario 3	16:00–21:00 (Whole event time)	2000.00	2013.17	**13.17**	–
	16:00–17:00	80.30	81.48	1.18	1.18
	17:00–18:00	385.66	386.65	0.99	2.17
	18:00–19:00	549.32	555.03	5.71	7.88
	19:00–20:00	510.77	515.32	4.55	12.43
	20:00–21:00	473.95	474.69	0.74	**13.17**

Table 7
Summary of data center performance in demand response environment across different scenarios.

Scenario	Performance			
	Electricity costs (Yuan)	Energy consumption (kWh)	ECD during DRE (kWh)	Cumulative ECD (kWh)
Scenario 1	33470.28	60655.82	–	–
Scenario 2	26890.15	49361.65	93.87	689.93
Scenario 3	**23357.49**	**37420.68**	**13.17**	**13.17**

values of 93.87 kWh and 13.17 kWh. The deficit rate decreased from 4.69 % to 0.66 %. Given that the DRE period spans five time slots, the electricity curtailment target and actual values in Scenario 2 fluctuate across these slots, with the gaps between them offsetting one another. Consequently, the ECD over the entire period fails to fully capture the significant advantage of Scenario 3 over Scenario 2. A clearer contrast effect is evident from the hourly cumulative ECD shown in Fig. 12(b). Compared to Scenario 2, in Scenario 3, the hourly cumulative ECD of Data Center 1 during the DRE period decreased by 98.09 %, from 689.93 kWh to 13.17 kWh.

Table 7 summarizes the performance of data centers across different scenarios in the demand response environment. As observed, compared to the conventional linear load model, the simplified nonlinear load model, based on the optimal energy consumption characteristics of data centers (adopted in Scenario 3) as proposed in this study, facilitates improved energy efficiency while demonstrating superior response quality during DREs.

6. Conclusions and future work

In the demand response environment, the conventional linear load model limits the enhancement of both energy efficiency and demand response quality in data centers. To address this issue, this study proposes a framework that extracts the optimal energy consumption characteristics of data centers from the solution to the energy optimization problem and, based on these characteristics, develops a reasonable simplified nonlinear load model for data centers. The foundational layer of the framework comprises fine-grained nonlinear models at the subsystem or component level within the data center, while the upper layer is a simplified nonlinear model constructed from the patterns identified in the solution to the energy optimization problem. This design ensures that the model derived from this framework not only more realistically reflects the energy consumption characteristics of the data center but also meets the model simplification requirements of large-scale optimization problems, such as demand response programming. In the simulation experiments, the simplified nonlinear load model proposed in this study, along with the conventional linear load model, is embedded into a typical demand response program involving two data centers. The performance of the data centers under various scenarios is evaluated and compared in terms of electricity cost, energy efficiency, and demand response quality, to validate the effectiveness of the modeling framework proposed in this study. The main conclusions and findings of this work include the following points:

(1) The complex data center system model, which incorporates fine-grained, highly nonlinear models at the subsystem and component levels, exhibits weakly nonlinear load-power characteristics following specific operational optimizations, such as the energy consumption optimization carried out in this study.
(2) The energy consumption optimization results, derived from a fine-grained nonlinear model considering multiple types of workloads and heuristic algorithms, suggest that, in the energy-optimal operational mode, latency-sensitive interactive workloads and latency-tolerant batch jobs should be co-deployed on servers, instead of being performed on separate, dedicated servers.
(3) The optimal energy consumption characteristics of the data center are defined by four modes, which can be represented as a piecewise function consisting of two linear regions and two nonlinear regions. Each region corresponds to optimal energy consumption under varying computational workloads. Moreover, the relationships in the nonlinear regions are reasonably simplified from cubic to quadratic forms. Specifically, in the linear regions, the model exhibits an R^2 value exceeding 0.9999 and an MRE value below 0.1404 %. In the quadratic regions, the R^2 value exceeds 0.9982, with a mean relative error under 0.6259 %.
(4) Compared to the conventional linear load model, the simplified nonlinear load model proposed in this study enables data centers to enhance both energy efficiency and demand response quality in a demand response environment. Specifically, the latter helps data centers achieve savings of 13.40 % to 30.21 % in electricity costs, reduce energy consumption by 24.19 % to 38.31 %, and lower the cumulative ECD during the DRE by 98.09 %. The actual outcomes depend on the specific operational modes of data centers employing the conventional linear model for demand response.

The modeling framework proposed in this study, which establishes load models for demand response based on the optimal energy consumption characteristics of data centers, is considered universally applicable. Through this framework, an increasing number of nonlinear relationships, which play a significant role in the energy consumption dynamics of data centers, can be integrated into the foundational nonlinear model layer. This integration updates our understanding of the optimal energy consumption characteristics of data centers. Furthermore, the simplified load models used for demand response can be refined based on these newly identified characteristics. This process enables data centers to achieve enhanced performance in demand response environments,

addressing the conflict between energy efficiency and demand response quality improvements resulting from model limitations.

Despite its contributions, the current study has several limitations. For instance, the optimal energy consumption characteristics of data centers identified in this work are derived based on the current model parameters. Although these parameters are considered reasonable at present, their applicability may be influenced by the ongoing technological advancements in various devices and systems within data centers. Consequently, these changes could lead to modifications in the optimal energy consumption characteristics. These deficiencies will be considered in our future work.

CRediT authorship contribution statement

Yongcheng Zhou: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Fanchao Wei:** Writing – original draft, Validation, Software, Methodology. **Shuangxiu Li:** Writing – review & editing, Investigation, Data curation. **Zhonghao Wang:** Writing – review & editing, Validation. **Jinfu Liu:** Supervision. **Daren Yu:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Conventional linear load model for data center demand response

As summarized in Table 1, to ensure the computational tractability of large-scale optimization problems, current data center load models used for demand response are largely based on simplified assumptions, resulting in linear models. In this section, such a conventional linear load model is established for use in the comparison experiments in Section 5. The total power consumption of the data center (P_{DC}) in this model is expressed as:

$$P_{\mathrm{DC}} = P_{\mathrm{cps}}^{\mathrm{DC}} + P_{\mathrm{PCU}}^{\mathrm{DC}} + P_{\mathrm{cl}}^{\mathrm{DC}}, \tag{A1}$$

where, $P_{\mathrm{cps}}^{\mathrm{DC}}$, $P_{\mathrm{PCU}}^{\mathrm{DC}}$, and $P_{\mathrm{cl}}^{\mathrm{DC}}$ denote the power consumption of the computing system, PCUs, and cooling system within the data center, respectively.

The main assumptions underlying the linear load modeling process include the following (these assumptions are widely applied in the modeling phases of data center demand response studies surveyed in Table 1 [16–32]):

(S1) The chip-level DVFS capability is ignored, and server chips are assumed to operate at their rated frequency at all times.
(S2) Fans in active servers are assumed to always run at their rated speed, thereby naturally satisfying the chip temperature threshold constraints.
(S3) Latency-sensitive interactive workloads and latency-tolerant batch jobs are assumed to be executed concurrently on all active servers, with priority given to interactive workloads.
(S4) Minimum number of servers is activated while ensuring QoS constraints are met.
(S5) The PCU in inactive modules remains in standby mode rather than being completely shut down, allowing the data center to quickly respond to increases in computational workloads.

The power of a single server (P_{s}) can be determined based on assumptions (S1), (S2), (S3), and (S4) using the following equation:

$$P_{\mathrm{s}} = P_{\mathrm{idle}} + u_{\mathrm{s}}\left(P_{\mathrm{peak}} - P_{\mathrm{idle}}\right), \tag{A2}$$

where, $P_{\mathrm{idle}} = 160.84$ W and $P_{\mathrm{peak}} = 652.84$ W denote the idle and peak power of a server operating with the chip at rated frequency and the fan at rated speed, respectively. $u_{\mathrm{s}} = \dfrac{a_{\mathrm{DC}} + b_{\mathrm{DC}}}{N_{\mathrm{s,on}}\mu}$ is the server utilization; $N_{\mathrm{s,on}}$ is the number of active servers. Although $N_{\mathrm{s,on}}$ is theoretically an integer variable, given the large number of servers and the relatively low power consumption of each individual server, it can be relaxed to a continuous variable for modeling purposes [21,30]. After relaxation, the minimum number of active servers required to meet QoS constraints, $N_{\mathrm{s,on}}$, can be determined as follows:

$$N_{\mathrm{s,on}} = \begin{cases} \dfrac{a_{\mathrm{DC}} D}{\mu D - 1}, & \text{for } b_{\mathrm{DC}}(\mu D - 1) \le a_{\mathrm{DC}} \\ \dfrac{a_{\mathrm{DC}} + b_{\mathrm{DC}}}{\mu}, & \text{for } b_{\mathrm{DC}}(\mu D - 1) > a_{\mathrm{DC}} \end{cases}; \tag{A3}$$

hence:

$$P_{\mathrm{cps}}^{\mathrm{DC}} = N_{\mathrm{s,on}} P_{\mathrm{s}} = \begin{cases} \dfrac{(\mu D-1)P_{\mathrm{peak}} + P_{\mathrm{idle}}}{(\mu D-1)\mu} a_{\mathrm{DC}} + \dfrac{P_{\mathrm{peak}} - P_{\mathrm{idle}}}{\mu} b_{\mathrm{DC}}, & \text{for } b_{\mathrm{DC}}(\mu D-1) \le a_{\mathrm{DC}} \\ \dfrac{P_{\mathrm{peak}}}{\mu} a_{\mathrm{DC}} + \dfrac{P_{\mathrm{peak}}}{\mu} b_{\mathrm{DC}}, & \text{for } b_{\mathrm{DC}}(\mu D-1) > a_{\mathrm{DC}} \end{cases}. \tag{A4}$$

Based on assumption (S5) and Eqs. (11) and (12), the power loss of the PCU equipment (PDU and UPS) in the data center can be calculated as follows:

$$\begin{aligned} P_{\mathrm{PCU}}^{\mathrm{DC}} &= N_{\mathrm{m}}\left[P_{\mathrm{UPS,idle}} + \left(\eta_{\mathrm{UPS}}+1\right)P_{\mathrm{PDU,idle}}\right] \\ &+ \left\lfloor \frac{N_{\mathrm{s,on}}}{N_{\mathrm{m}}} \right\rfloor \left[\eta_{\mathrm{PDU}}\left(\eta_{\mathrm{UPS}}+1\right)\left(N_{\mathrm{m,s}}P_{\mathrm{s}}\right)^2 + \eta_{\mathrm{UPS}} N_{\mathrm{m,s}} P_{\mathrm{s}}\right] \\ &+ \eta_{\mathrm{PDU}}\left(\eta_{\mathrm{UPS}}+1\right)\left[\left(N_{\mathrm{s,on}} - N_{\mathrm{m,s}}\left\lfloor \frac{N_{\mathrm{s,on}}}{N_{\mathrm{m}}} \right\rfloor\right)P_{\mathrm{s}}\right]^2 \\ &+ \eta_{\mathrm{UPS}}\left(N_{\mathrm{s,on}} - N_{\mathrm{m,s}}\left\lfloor \frac{N_{\mathrm{s,on}}}{N_{\mathrm{m}}} \right\rfloor\right)P_{\mathrm{s}}. \end{aligned} \tag{A5}$$

Fig. A1 illustrates the relationship between the power of the computing system in the data center, calculated using Eq. (A4) under various operating conditions, and the power of the PCU equipment calculated using Eq. (A5). Evidently, a linear model based on the power of the computing system can accurately estimate the power of the PCU equipment (with $R^2 = 0.9995$), as shown in Eq. (A6):

$$P_{\mathrm{PCU}}^{\mathrm{DC}} = P_{\mathrm{PCU,idle}}^{\mathrm{DC}} + \eta_{\mathrm{PCU}} P_{\mathrm{cps}}^{\mathrm{DC}}, \tag{A6}$$

where, $P_{\mathrm{PCU,idle}}^{\mathrm{DC}} = N_{\mathrm{m}}\left[P_{\mathrm{UPS,idle}} + \left(\eta_{\mathrm{UPS}}+1\right)P_{\mathrm{PDU,idle}}\right] = 80467.87$ W denotes the standby power consumption of the PCU equipment in the data

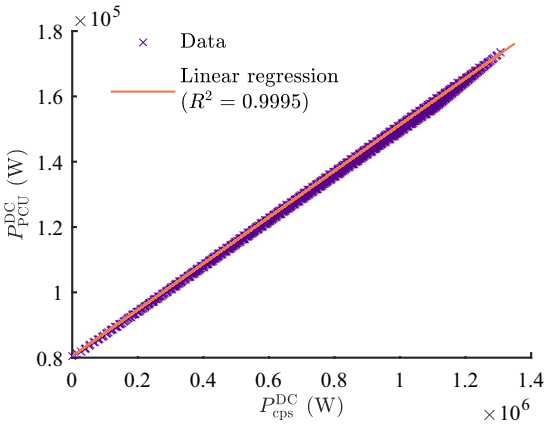


Fig. A1. The relationship between the PCU power and the computing system power in the data center.

center; $\eta_{\mathrm{PCU}} = 0.07091$ represents the equivalent efficiency coefficient of the PCU linear power model.

For the modeling of the cooling system power consumption, the influence of supply air temperature on system efficiency is disregarded, and the cooling power ($P_{\mathrm{cl}}^{\mathrm{DC}}$) is calculated using a fixed CoP value, that is:

$$P_{\mathrm{cl}}^{\mathrm{DC}} = \frac{P_{\mathrm{ncl}}^{\mathrm{DC}}}{CoP_{\mathrm{fix}}} = \frac{P_{\mathrm{cps}}^{\mathrm{DC}} + P_{\mathrm{PCU}}^{\mathrm{DC}}}{CoP_{\mathrm{fix}}}, \tag{A7}$$

where, $P_{\mathrm{ncl}}^{\mathrm{DC}}$ represents the power of the non-cooling systems in the data center. CoP_{fix} is the fixed CoP value of the cooling system. Specifically, in this study, CoP_{fix} is chosen as the median of the CoP values corresponding to the ASHRAE-recommended cooling air supply temperature range (18 °C–27 °C) [52], as calculated using Eq. (15), which yields $CoP_{\mathrm{fix}} = 4.0562$.

Substituting Eqs. (A4), (A6), and (A7) into Eq. (A1), Eq. (A1) can be rewritten as:

$$P_{\mathrm{DC}} = \begin{cases} p_1 + p_2 a_{\mathrm{DC}} + p_3 b_{\mathrm{DC}}, & \text{for } b_{\mathrm{DC}}(\mu D - 1) \le a_{\mathrm{DC}} \\ p_1 + p_4 a_{\mathrm{DC}} + p_4 b_{\mathrm{DC}}, & \text{for } b_{\mathrm{DC}}(\mu D - 1) > a_{\mathrm{DC}} \end{cases}; \tag{A8}$$

where, $p_1 = \dfrac{P_{\mathrm{PCU,idle}}^{\mathrm{DC}}\left(CoP_{\mathrm{fix}} + 1\right)}{CoP_{\mathrm{fix}}}$,

$p_2 = \dfrac{\left(1 + \eta_{\mathrm{PCU}}\right)\left(CoP_{\mathrm{fix}} + 1\right)\left[(\mu D - 1)P_{\mathrm{peak}} + P_{\mathrm{idle}}\right]}{CoP_{\mathrm{fix}}(\mu D - 1)\mu}$,

$p_3 = \dfrac{\left(1 + \eta_{\mathrm{PCU}}\right)\left(CoP_{\mathrm{fix}} + 1\right)\left(P_{\mathrm{peak}} - P_{\mathrm{idle}}\right)}{CoP_{\mathrm{fix}}\mu}$,

and $p_4 = \dfrac{\left(1 + \eta_{\mathrm{PCU}}\right)\left(CoP_{\mathrm{fix}} + 1\right)P_{\mathrm{peak}}}{CoP_{\mathrm{fix}}\mu}$ are all model parameters.

Data availability

Data will be made available on request.

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